

Beneath the VIX: Probability, Severity, and Uncertainty Shocks*

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Abstract

Implied stock-market volatility is widely used to identify uncertainty shocks and discipline models of their macroeconomic effects. This paper gives distributional content to the option-price variation underlying these exercises. A normalized at-the-money forward put price, the Downside Protection Premium (DPP), closely tracks the VIX and preserves the principal macroeconomic responses in the specifications examined. Its price and local strike slope separate risk-neutral shortfall probability from conditional severity. Severity accounts for almost all monthly premium comovement and 91.5 percent of the recession log premium. Conditional gains also rise substantially, while conditional losses remain larger in phase averages. Despite this severity dominance, quotations across strikes frequently exclude a common-scale explanation, conditional on the maintained forward and premium construction. Realized-volatility and future-volatility-news innovations both raise protection costs primarily through severity, yet produce different estimated employment responses in common impact-severity units; simultaneous significance depends on the inference formulation. Together, these findings supply joint empirical restrictions on the connection between uncertainty, option prices, and economic activity.

JEL Codes: E32, G12, G13

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1 INTRODUCTION

Volatility indices such as the VIX and VXO are widely used proxies for market uncertainty in macroeconomics. These measures provide observables from which uncertainty innovations are identified and targets used to discipline quantitative macroeconomic models of uncertainty transmission. Among other results, a VXO or VIX shock has been shown to produce a sharp drop and rebound in output and employment within about six months (Bloom, 2009); lower output, consumption, investment, and hours together (Basu and Bundick, 2017); raise unemployment and lower inflation, much like a negative aggregate demand shock (Leduc and Liu, 2016); raise unemployment by substantially more in recessions than in expansions (Caggiano et al., 2014); and have larger contractionary effects when the zero lower bound binds (Caggiano et al., 2017). However, Berger et al. (2020) find that the contraction is associated with realized-volatility shocks, while news about future volatility has little effect on activity. This paper similarly looks beneath the volatility index. I construct the price of a one-month at-the-money forward put, normalized by the index level, and show that this downside protection premium moves almost perfectly with the VIX. Its price equals the product of the risk-neutral probability of a shortfall and the expected severity of that shortfall. I show that severity accounts for nearly all of the premium's movements. Moreover, I document that innovations to realized volatility and news about future volatility both raise protection costs mainly through severity, yet they have different estimated effects on employment.

I begin with the price of an at-the-money forward put divided by the current index value, which I call the Downside Protection Premium (DPP). This price measures the cost of protection against the index finishing below its forward value. Its monthly correlation with the VIX is 0.992, and substituting the DPP for the VIX or VXO as a proxy for uncertainty preserves the principal macroeconomic responses well-known settings Bloom (2009); Leduc and Liu (2016); Basu and Bundick (2017). The DPP, therefore, captures the market-price variation underlying familiar macroeconomic uncertainty exercises.

The DPP permits a direct interpretation of that variation. After adjusting for carrying costs, the premium equals the risk-neutral probability of finishing below the forward (i.e., probability of shortfall) multiplied by the expected proportional shortfall conditional (i.e., conditional severity) on doing so. The DPP pins down only the product, but puts with nearby strikes can be used to separate the two because raising the strike slightly increases the put's payoff only when the index finishes below it, so the resulting change in price reveals the probability of a shortfall. Thus, simple put prices do more than track the VIX, these prices also support a familiar insurance interpretation of its movements. When the cost of protection rises, is a loss more likely, or would it be more severe?

Thus, the paper is able to address three questions. First, which component accounts for movements in the protection premium? Second, can that composition be explained by a change in scale alone, that is, by a distribution of index outcomes that simply widens or narrows around the forward without changing its shape? Such a pure widening is a natural benchmark because it keeps the shortfall probability fixed and moves the premium entirely through severity, while also fixing how the prices of puts at other strikes move relative to the at-the-money price. Third, do volatility innovations that raise the premium through a similar mix of probability and severity have similar macroeconomic consequences? Together, these questions turn the decomposition into empirical restrictions on the connection between uncertainty, option prices, and economic activity.

My first finding is that conditional severity accounts for almost all monthly premium comovement and most of the recession premium. Using daily SPX option quotations from January 1990 through July 2026, I estimate a monthly severity loading of 1.032 and a complementary probability loading of -0.032 . The probability loading's 95 percent interval, $[-0.071, 0.006]$, includes zero, and probability changes are substantially less variable than severity changes. Severity accounts for 91.5 percent of the mean log premium difference between recessions and expansions, although the recession probability contribution is positive. These conclusions persist across time periods,

recession comparisons, and alternative component-recovery methods. Quarterly-horizon options give the same broad pattern of severity dominance.

The interpretation is two-sided due to put-call parity. Because the forward is the market's price-weighted average of where the index will finish, expected shortfalls below the forward must be offset by expected gains above it. By put-call parity, the adjusted premium therefore also equals the probability of finishing at or above the forward multiplied by the expected gain conditional on doing so, which I call conditional gain severity. The monthly gain-severity loading is 0.977, and conditional gains rise substantially during recessions. Higher protection costs thus accompany larger conditional movements on both sides of the forward, rather than an exclusively downside movement. The probability determines the relative size of these movements. The shortfall probability averages 0.423 in expansions and 0.444 in recessions—shortfalls are priced as less likely than gains, so for expected losses and gains to balance, a shortfall must be larger than a gain when it occurs. The conditional loss is roughly 36 percent larger than the conditional gain in expansions and 25 percent larger in recessions. The priced distribution thus describes frequent modest gains and less frequent, larger losses. This asymmetry is information that the dominance of severity does not convey. A lognormal distribution, the benchmark behind Black–Scholes, also moves its premium almost entirely through severity; doubling the premium from two to four percent raises severity from 3.92 to 7.69 percent while the probability barely moves, from 0.51 to 0.52. But it implies the opposite asymmetry. Its outcomes are skewed to the right, with a long upper tail, so the median lies below the mean and more than half the outcomes fall short of the forward. A lognormal therefore prices shortfalls as more likely than gains but smaller, whereas the data price them as less likely but larger. Thus, matching the volatility index, or even the dominance of severity, is not sufficient. The data clearly want a model that prices shortfalls as less likely than gains but larger when they occur, which rules out a lognormal distribution.

The second finding is that the dominance of severity does not mean that option prices simply reflect a widening distribution. If the distribution of index outcomes only widens or narrows around the forward without changing its shape, the shortfall probability stays fixed and the premium moves one-for-one with severity. Such a widening has a testable implication. Once the at-the-money premium reveals how wide the distribution is on a given date, the prices of puts at every other strike follow mechanically. Within each month, I therefore ask whether any single shape, widened or narrowed day by day, can match the quoted prices across strikes to within their bid–ask spreads. In 352 of 405 months, no such shape exists. A weaker version asks only whether the shape of losses, given that a shortfall occurs, stays fixed, allowing the shortfall probability and the distribution of gains to change freely from day to day. Even this weaker version fails in 112 months, or 27.7 percent of the sample. These counts take the forward and premium construction as given; they are not statistical rejection rates, and some of the discrepancies are small. Nonetheless it shows that the at-the-money price can move almost entirely through severity while the shape of the priced distribution changes in ways that a pure widening cannot explain. Thus, for a quantitative model disciplined by implied volatility, matching the volatility index is not sufficient; its equilibrium option prices must also generate changes in the shape of the priced distribution, whether through nonlinear payoffs, state-dependent valuation, or the uncertainty shock itself.

The third finding connects the composition of protection prices to differences in macroeconomic responses. Berger et al. (2020) distinguish innovations to realized volatility from news about future volatility and find contrasting responses of economic activity. I extend this comparison to the probability and severity components of the protection premium. Both innovations raise protection costs almost entirely through conditional severity, yet manufacturing employment falls more after the realized-volatility innovation. To put these employment responses on a common scale, I normalize each innovation to raise conditional severity by one percent on impact. Under this normalization, the estimated manufacturing employment response twelve months later is about 0.065 percentage

points lower after a realized-volatility innovation than after volatility news. The employment contrast therefore survives equalizing the impact severity response. This comparison supplies a joint empirical target for models of uncertainty transmission—both innovations raise the severity of priced shortfalls, but the estimated employment responses differ.

1.1 RELATION TO EXISTING LITERATURE The paper primarily contributes to the macroeconomic literature that uses implied volatility to identify uncertainty shocks and discipline models of their effects (Bloom, 2009; Leduc and Liu, 2016; Basu and Bundick, 2017). I show that replacing the VIX or VXO with the DPP in the empirical specifications of these papers preserves their main conclusions about how uncertainty shocks affect economic activity. Unlike the VIX, the DPP can be decomposed which I exploit to show that most of the option-price variation attributable to severity rather than probability.

Other papers decompose uncertainty into components with different economic effects. Segal et al. (2015) find that good uncertainty about macroeconomic growth predicts higher future activity and asset valuations, whereas bad uncertainty predicts lower ones. Forni et al. (2025) find that widening the left tail of the GDP-growth forecast distribution is contractionary, whereas widening the right tail is mildly expansionary. My decomposition differs in both object and components, and provides additional facts that structural models must explain.

The closest comparison is Berger et al. (2020). Their distinction between realized-volatility innovations and news about future volatility has different implications for economic activity. I add the priced composition of the associated protection-price responses and compare activity in common impact-severity and protection-premium units. Both innovations produce severity-dominated responses, yet their estimated employment responses differ in these units. The additional evidence directs structural interpretation toward the origins, subsequent paths, and other equilibrium effects of the innovations. Related measurement work connects at-the-money option statistics to implied volatility. Dew-Becker and Giglio (2023) use at-the-money implied volatility in their analysis of cross-sectional uncertainty. Dotsis (2024) constructs an option-implied absolute-deviation index and studies its relationship with the VIX and the predictive content of their spread for equity returns. The present paper connects the at-the-money protection price to macroeconomic uncertainty identification and model evaluation. It examines the composition of the price movements underlying standard uncertainty exercises, their compatibility with common-scale restrictions, and their association with economic activity. The contribution consequently extends beyond the close correspondence between an at-the-money statistic and the VIX.

The distributional analysis is also related to work on option-implied skewness. Dew-Becker (2024) finds that aggregate option-implied skewness is approximately acyclical, while firm-level skewness is procyclical. The local probability gap and conditional loss–gain ratio studied here are distinct from a global standardized third moment. On identical quarter-end observations, the recession change in the premium-adjusted local probability gap is larger for 30-day than for quarterly-horizon options. This horizon dependence and the joint quotation restrictions address properties of the priced distribution that global skewness alone does not determine. The analysis does not reinterpret the local gap as a new estimator of global skewness or claim that its behavior overturns the aggregate skewness finding.

Options have also been used to assess disaster mechanisms and the pricing of adverse states. Backus et al. (2011) infer disaster implications from index options, while Seo and Wachter (2019) study option prices in a model with stochastic disaster risk. Bekaert and Engstrom (2017) model non-Gaussian fundamentals and valuation, and Chabi-Yo and Loudis (2024) decompose conditional risk premia across downside, upside, and central regions. Culp et al. (2021) relate option-implied spreads to risk-neutral probabilities of finishing below the strike and conditional losses given that event, and use these measures to study option risk premia. The present exercise con-

cerns the composition of protection costs and their connection to empirical uncertainty innovations; it does not separately recover physical probabilities and risk premia. To the best of my knowledge, this paper is the first to jointly document three features of the option-price variation underlying macroeconomic uncertainty exercises: conditional shortfall severity accounts for nearly all monthly protection-premium comovement; quotations exclude within-month common scale in most months; and realized-volatility innovations and volatility news have different estimated employment responses even when normalized to the same impact.

2 THE DOWNSIDE PROTECTION PREMIUM

Consider an at-the-money forward (AtM-forward) put option, defined by a strike K equal to the no-arbitrage forward price $F_{t,t+\tau}$, where the forward definition accounts for the cost of carrying the underlying to expiration. Let r_t and d_t denote the continuously compounded risk-free rate and dividend yield, held constant over the option horizon. Then $F_{t,t+\tau} = S_t \exp[(r_t - d_t)\tau]$, where S_t is the current underlying price and τ is time to maturity in years. Let $P_t^{\text{AtM}}(\tau)$ denote the price of this European put. I define the Downside Protection Premium (DPP) as its price divided by the current underlying price¹

$$\chi_t(\tau) \equiv \frac{P_t^{\text{AtM}}(\tau)}{S_t} \quad (1)$$

For much of the analysis, I focus on 30 calendar days, $\tau = 30/365$, and write $\chi_t^{30} \equiv \chi_t(30/365)$ to match the horizon of the VIX.

The DPP measures the cost of downside protection per dollar of current equity exposure and has an intuitive insurance interpretation—it is the cost of one month of protection against the index finishing below its forward price per dollar of current equity exposure. An investor holding \$100 of the index at date t pays $100\chi_t^{30}$ dollars for the corresponding put. Equivalently, the premium is $100\chi_t^{30}$ percent of current equity exposure. Moreover, the ratio is dimensionless, allowing comparisons across time and assets.

2.1 CONSTRUCTING A SYNTHETIC ATM-FORWARD PUT PRICE I use daily end-of-day SPX option quotes from DeltaNeutral (HistoricalOptionData.com) to construct a daily time series of the Downside Protection Premium, χ_t^{30} . The data contain puts and calls across a range of strikes and expirations from January 1990 through July 2026. I impose neither a minimum trading volume nor a maximum relative bid–ask spread. I price options at the midpoint of the bid–ask spread and retain quotes with a nonnegative bid, a strictly positive ask, and an ask no lower than the bid. For put–call parity and additional calculations below, I match put and call contracts within the same recorded time frame by strike, expiration, and identified contract family and settlement terms. Note that SPX options have European exercise, avoiding the need for an early-exercise adjustment. Appendix A.1 provides additional details and documents a substantial increase in volume around 2005, with put volume exceeding calls, and with most put volume occurring near-the-money.

In order to estimate a synthetic put price at the forward strike and a constant 30-day horizon, I use two linear interpolations designed to be simple by construction. The first targets the forward strike separately at each maturity bracketing the 30-day horizon, using $F_t(T; K) = K + e^{r_t T} [C_t(T, K) - P_t(T, K)]$. For both short and long durations, I take the median implied forward from up to three matched strikes closest to the contemporaneous index level and denote it by $F_t(T)$. The current implementation uses a daily three-month Treasury bill rate, con-

¹The close empirical DPP–VIX correspondence established below also holds with simpler at-the-money constructions. Holding quote dates, expirations, and interpolation in time fixed, interpolating put prices to the current index level, $K = S_t$, yields a nearly identical monthly relationship with the VIX. Even using the quoted put strike closest to spot without interpolation across strikes works nearly as well. Thus, the close monthly relationship does not require the forward adjustment. It is retained throughout for the exact forward-centered probability–severity decomposition and loss–gain identities below.

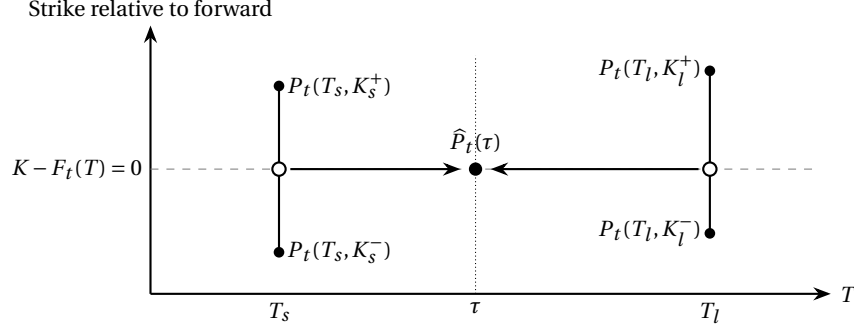


Figure 1: Constructing the synthetic AtM-forward put price. At each bracketing maturity, the two observed put prices (solid endpoints) are interpolated across strikes to obtain a price at the forward (open circles). Interpolation between these two prices gives the estimate at the target horizon (central solid circle). Vertical positions represent strike distances from the maturity-specific forward.

verted to continuously compounded units with past rates carried forward for at most three calendar days. Using parity to infer the forward incorporates the pricing of dividends without requiring a separate dividend series; see van Binsbergen et al. (2012). Let $K^- < F_t(T) < K^+$ be the nearest eligible put strikes below and above the targeted forward. The interpolated AtM-forward put price for that expiration is therefore

$$\hat{P}_t^{\text{AtM}}(T) = \left(\frac{K^+ - F_t(T)}{K^+ - K^-} \right) P_t(T, K^-) + \left(\frac{F_t(T) - K^-}{K^+ - K^-} \right) P_t(T, K^+)$$

I limit the total width of the strike bracket to five percent of the forward $(K^+ - K^-)/F_t(T) \leq 0.05$; this restriction drops 6 days.

The second interpolation pins down maturity by selecting eligible maturities immediately above and below the target duration, $T_s < \tau < T_l$. The synthetic price is the combination

$$\hat{P}_t(\tau) = \left(\frac{T_l - \tau}{T_l - T_s} \right) \hat{P}_t^{\text{AtM}}(T_s) + \left(\frac{\tau - T_s}{T_l - T_s} \right) \hat{P}_t^{\text{AtM}}(T_l) \quad (2)$$

I impose no additional restriction on the distance between the bracketing maturities. If the required strike or maturity brackets are unavailable, the daily observation remains missing. The procedure neither extrapolates nor fills missing dates. The construction is intentionally simple with the final price using a convex combination of four put prices closest to the desired AtM strike and 30-day duration.

The estimated Downside Protection Premium, $\hat{\chi}_t^{30}$, is available on 8,825 trading days from January 2, 1990 through July 31, 2026, covering 95.80 percent of the 9,212 regular trading sessions in the sample.² I aggregate the daily DPP to monthly frequency by taking the arithmetic average of available daily observations within each month. The resulting series contains 439 monthly observations with no missing months from January 1990 through July 2026. Panel A of Table 1 reports monthly summary statistics. The mean monthly DPP is 1.88 percent, corresponding to a premium of approximately \$1.88 for 30 days of protection per \$100 of current equity exposure. The median monthly premium is substantially lower than the mean at 1.68 percent. Hence, there is substantial right skewness (2.039). The standard deviation of the monthly series is 0.77 percentage points, compared with 0.81 percentage points for the daily series; monthly averaging leaves the mean premium nearly unchanged. The monthly

²Of the 387 sessions without a constructed premium, two have no source SPX quotations, 379 lack a usable maturity bracket for the 30-day construction, and six are due to the 5 percent strike-bracket restriction. Of these 379 maturity failures, 359 occur between the 14th and 21st of the month and coincide with dates when the next standard monthly expiration is more than 30 days away. The remaining 20 occur between September 18 and October 15, 1992.

series is persistent with a first-order autocorrelation of 0.853.

2.2 CONNECTION TO THE VIX AND UNCERTAINTY SHOCKS Panels C and D of Table 1 establish the empirical connection motivating the DPP for the subsequent analysis of uncertainty shocks. *The DPP and the VIX are almost perfectly correlated at every sampling frequency and in both levels and changes.* The Pearson correlation is 0.990 across 8,822 matched daily observations and 0.992 across 439 monthly averages; the Spearman correlations are nearly as high (0.984 and 0.988), which, together with the expansion-month correlation reported below, indicates that the association extends beyond crisis observations. Correlations in monthly log changes are also quite high at 0.984, so the relationship is not an artifact of common trends or shared persistence in levels. The comovement is stable over time and across business-cycle phases. Recession months are more highly correlated 0.995 than expansions 0.988, so the two measures are more strongly coupled when uncertainty is high. Rolling 24-month correlations never fall below 0.950 in levels or 0.921 in log changes, with medians of 0.991 and 0.987. The one visibly lower reading is the 12-month minimum of 0.716 in levels.

Panel B regresses the DPP in basis points of exposure on the VIX, which yields a slope of 10.33 (HAC standard error 0.12) in levels and 10.13 (0.14) in first differences, with R^2 of 0.985 and 0.980. Thus, a one-point increase in the VIX is associated with an increase in the cost of one month of protection of roughly 10 basis points of exposure, and the near-identical slopes support an approximately linear relationship in levels and changes. The small negative intercept in levels (−13.2 basis points) and zero counterpart in changes (−0.01) are consistent with an approximately linear relationship.

Table 1: The 30-Day DPP: Summary Statistics and Relationship with the VIX

A. Summary statistics							B. Regression analysis				
Mean	SD	Min	Med.	Max	Skew.	AR(1)	Series	Const.	Slope	R^2	RMSE
1.88	0.77	0.81	1.68	6.35	2.04	0.85	Levels	−13.2 (2.21)	10.33 (0.12)	0.985	9.45
							Changes	−0.01 (0.15)	10.13 (0.14)	0.980	5.89

C. Correlations with VIX				D. Rolling correlations				
Comparison	Obs.	Pearson	Spear.	Series	Window	Min	Med.	Max
Daily levels	8,822	0.990	0.984	Levels	12	0.716	0.991	1.000
Monthly levels	439	0.992	0.988	Log chg.	12	0.856	0.987	0.999
Monthly log chg.	438	0.984	0.979	Levels	24	0.950	0.991	0.999
Recessions	36	0.995	0.971	Log chg.	24	0.921	0.987	0.996
Expansions	403	0.988	0.986					

Note: Sample 1990-01-02 to 2026-07-31; 8,825 daily observations averaged into 439 calendar months. Panel A reports monthly summary statistics for $100\chi_t^{30}$, the 30-day put premium as a percent of exposure; skewness is bias-adjusted and AR(1) is the adjacent-month autocorrelation. Panels B–D align DPP and the VIX on 8,822 common daily dates before averaging (3 DPP dates lack VIX quotes). Panel B estimates $10,000\chi_t^{30} = \alpha + \beta VIX_t + u_t$ and its first difference. Intercepts and RMSEs are in basis points of exposure, and slopes are in basis points of exposure per VIX point; parentheses give Bartlett HAC standard errors (12 lags). Panel C reports unitless correlations and splits months by NBER phase, with recessions running from the month after a peak through the trough. Panel D uses overlapping windows of consecutive months.

Table 1 shows that the DPP and VIX share most of their observed time-series variation. Appendix A.2 examines the implications of substituting the DPP for the VIX/VXO in three influential papers, Bloom (2009); Leduc and Liu (2016); Basu and Bundick (2017). Swapping the VIX/VXO for the DPP preserves the principal macroeconomic responses to uncertainty shocks in these papers, although event dates and some inference classifications based

on subjective metrics differ (see, Appendix A.2). The aggregate time-series information in the DPP thus closely matches that of the VIX/VXO.

2.3 INTUITION Why is the DPP so strongly temporally correlated with the VIX? The close correspondence has a natural historical antecedent. The original VIX, developed by Whaley (1993), combined the implied volatilities of eight near-the-money S&P 100 options to represent a 30-day AtM option. As noted above, the DPP construction uses four near-the-money put prices and shares this emphasis on near-the-money options.³ However, the VIX was redesigned in 2003 to extend to a broader range of strikes to capture more tail behavior (Whaley, 2009). The question is how much this broader strike coverage changes the fluctuations already captured by an AtM price.

Black–Scholes formulas offer intuition. For an exact forward-AtM put, its implied volatility $\sigma_{\text{AtM},t}(\tau)$ satisfies

$$\chi_t^\tau = e^{-d_t\tau} \left[2\Phi\left(\frac{\sigma_{\text{AtM},t}(\tau)\sqrt{\tau}}{2}\right) - 1 \right] \approx \frac{e^{-d_t\tau}}{\sqrt{2\pi}} \sigma_{\text{AtM},t}(\tau)\sqrt{\tau}$$

where Φ is the standard normal distribution function and d_t is the dividend yield. This is an implied-volatility representation of the option price, not an assumption that the actual risk-neutral distribution is lognormal. At $\tau = 30/365$ and $d_t = 0.02$, this gives $100 \cdot \sigma_{\text{AtM},t} \approx 876\chi_t^{30}$. Thus, the DPP is approximately proportional to AtM implied volatility, and one can assess the empirical departure from proportionality through the relative gap, $R_t = \text{VIX}_t/(876\chi_t^{30})$, and $\Delta \log \text{VIX}_t = \Delta \log \chi_t^{30} + \Delta \log R_t$. Using matched monthly levels, the median ratio is 1.187. The standard deviation of $\Delta \log R_t$ is 0.0312, compared with 0.1614 for $\Delta \log \text{VIX}_t$. The relative gap thus changes much less than the VIX itself, consistent with a dominant common component. This evidence, along with Figure 2 (which plots the approximation $876\chi_t^{30}$), supports approximate proportionality.⁴

Beyond the Black-Scholes algebra, there is a common-scale interpretation that will be revisited throughout the rest of the paper. Let $X = S_{t+\tau}/F_{t,t+\tau}$ denote the forward-normalized terminal index value, so that $E_t^Q[X] = 1$ under the maintained carrying-cost assumptions. The DPP and an idealized single-maturity log-contract volatility measure are then determined by expectations of functions of the single variable X

$$\chi_t^\tau = \frac{e^{-d_t\tau}}{2} E_t^Q[|X - 1|] \quad \nu_t(\tau)^2 = -\frac{2}{\tau} E_t^Q[\log X]$$

where $\nu_t(\tau)$ denotes log-contract volatility, distinct from the observed VIX with its discrete strikes and maturity interpolation. The first identity expresses the DPP as one half of the discounted risk-neutral mean absolute deviation of X . Equivalently, by put–call parity, twice the DPP is the normalized at-the-money-forward straddle price. The second is the log-contract representation, where $\nu_t(\tau)^2$ equals $2/\tau$ times the risk-neutral entropy $\log E_t^Q[X] - E_t^Q[\log X]$. The latter is recovered from a continuum of strikes at a single maturity (Carr and Wu, 2009). To isolate the role of scale, hold carrying costs fixed and write $X = 1 + s_t Z$, where Z has a fixed risk-neutral distribution with bounded support, mean zero, and unit variance, and $s_t > 0$ is small enough that $X > 0$ almost surely. The DPP is then *exactly* linear in s_t $\chi_t^\tau = 0.5 e^{-d_t\tau} s_t E^Q[|Z|]$. The log-contract volatility measure is linear only to first order. Expanding the log contract as s_t approaches 0,

$$\nu_t(\tau) = \frac{s_t}{\sqrt{\tau}} \left(1 - \frac{\mu_3}{3} s_t + O(s_t^2) \right) \quad \mu_3 \equiv E^Q[Z^3]$$

Several points are noteworthy. First, a pure change in scale moves both measures proportionally to first order,

³The original VIX, subsequently designated VXO, used American OEX options, defined AtM relative to the current index, and interpolated implied volatilities. The DPP uses European SPX options, defines AtM relative to the forward, and interpolates prices.

⁴Related evidence appears in Dotsis (2024), whose option-implied absolute-deviation index has a monthly level correlation of 0.990 with VIX over 1996–2021.

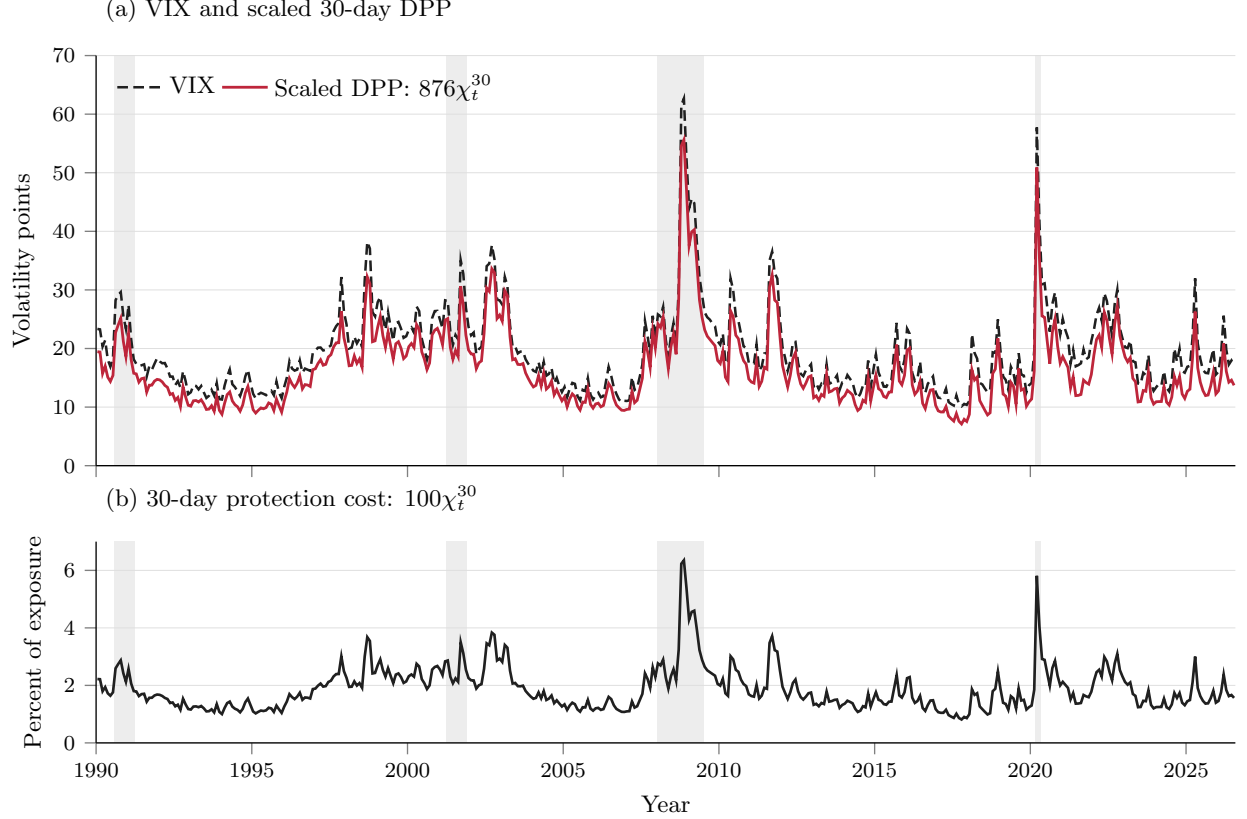


Figure 2: Monthly 30-day downside protection premium (DPP) and VIX. Panel (a) compares the VIX with the fixed benchmark $876\chi_t^{30}$ in volatility points, where 876 is the Black-Scholes approximation factor. Panel (b) reports the same DPP in percent of current equity exposure, $100\chi_t^{30}$. Both series are arithmetic averages of daily observations. NBER recession months are gray bars. The daily sample is 1990-01-02 through 2026-07-31.

whatever the shape of Z . Second, skewness enters the first correction to proportionality, but this correction depends jointly on μ_3 and s_t . The ratio of DPP to log-contract volatility therefore varies with scale even when skewness is fixed, and higher moments enter at subsequent orders. This is the sense in which the two measures share a first-order loading on scale while remaining distinct.⁵ Finally, this is a static comparison, at a single maturity, under the maintained risk-neutral pricing assumptions. While it does not specify dynamics for s_t , there is a conditional dynamic implication. Because the two measures are approximately linear in s_t , persistent variation in dispersion can generate persistent and highly correlated time series when the higher-order corrections are small. A common scale component is therefore one explanation for the dominant shared variation in the VIX and DPP. Their high correlation does not establish that the shape of the priced distribution is unchanged; distributional shape and valuation can also vary. The next section uses the local strike slope of put prices to distinguish the risk-neutral probability of a shortfall from its conditional severity. Put-call parity also gives the premium an upside representation, so this interpretation concerns conditional movements on both sides of the forward. The price and probability contain the same local information as the AtM price and implied-volatility slope, while the decomposition expresses that information in probability and conditional-magnitude units.

⁵This mechanism is consistent with the prominent level factor in implied-volatility surface variation documented by Cont and da Fonseca (2002).

3 DECOMPOSING THE DPP

The preceding section establishes that the DPP closely tracks the VIX and preserves the principal macroeconomic responses in the specifications examined. I now recover its probability and severity components and establish two results: conditional severity accounts for nearly all monthly premium comovement and most of the recession premium, but a *common* expansion of the priced distribution does not generally account for quotations across strikes.

3.1 THE PROBABILITY–SEVERITY DECOMPOSITION Let $\tilde{R}_t^\tau = S_{t+\tau}/F_{t,t+\tau}$ denote the terminal index value relative to its forward, and let $D_t^\tau = e^{-d_t^\tau}$ denote the carry factor. Under the maintained pricing assumptions, $\mathbb{E}_t^\mathbb{Q}[\tilde{R}_t^\tau] = 1$, and the AtM-forward put price over spot satisfies

$$\chi_t^\tau = D_t^\tau \underbrace{\mathbb{Q}_t(\tilde{R}_t^\tau < 1)}_{q_t^\tau} \underbrace{\mathbb{E}_t^\mathbb{Q}[1 - \tilde{R}_t^\tau | \tilde{R}_t^\tau < 1]}_{m_t^\tau}. \quad (3)$$

Here q_t^τ is shortfall probability and m_t^τ is conditional shortfall severity. A shortfall means finishing below the forward, rather than crossing a fixed disaster threshold. Both components are risk-neutral quantities. Removing carry gives the adjusted premium $a_t^\tau \equiv \chi_t^\tau / D_t^\tau = q_t^\tau m_t^\tau$. Its price alone identifies the product; a local option-price slope will provide the additional information needed to recover the components.

The product identity gives a direct interpretation of premium changes. Along a differentiable distributional path with a changing premium,

$$\eta_q^\tau \equiv \frac{d \log q^\tau}{d \log a^\tau} \quad \eta_m^\tau \equiv \frac{d \log m^\tau}{d \log a^\tau} \quad \eta_q^\tau + \eta_m^\tau = 1 \quad (4)$$

A unit severity elasticity means that conditional losses rise proportionately with the premium while shortfall probability is unchanged. The forward-mean restriction also equates expected losses and gains

$$a_t^\tau = q_t^\tau m_t^\tau = (1 - q_t^\tau) m_t^{+, \tau} \quad \frac{m_t^\tau}{m_t^{+, \tau}} = \frac{1 - q_t^\tau}{q_t^\tau} \quad (5)$$

where $m_t^{+, \tau}$ is the expected proportional gain conditional on finishing at or above the forward. Thus $q_t^\tau < 1/2$ implies larger conditional priced losses than gains. This asymmetry expresses the same probability information in conditional-loss units; it is distinct from third-moment skewness.

3.2 SEVERITY DOMINANCE IN SCALE BENCHMARKS A distribution that expands around the forward without changing shape provides the simplest explanation of severity dominance.

Proposition 1 (Scale and conditional severity). *Suppose $\tilde{R}_t^\tau = 1 + s_t Z$, where $s_t > 0$, Z has a fixed risk-neutral distribution with mean zero and unit variance, and $1 + s_t Z > 0$ almost surely at each date considered. Then*

$$q_t^\tau = \Pr(Z < 0), \quad m_t^\tau = s_t \mathbb{E}[-Z | Z < 0], \quad a_t^\tau = s_t \mathbb{E}[(-Z)^+]. \quad (6)$$

Shortfall probability is constant, and all proportional premium variation is attributable to conditional severity.

Proof. Since $s_t > 0$, the events $\tilde{R}_t^\tau < 1$ and $Z < 0$ coincide, and $1 - \tilde{R}_t^\tau = -s_t Z$. Taking the probability, conditional expectation, and expected put payoff gives the three equalities. $\square$

The distribution of Z may be asymmetric, so the constant probability need not equal one-half. The restriction is fixed shape as scale changes. Severity can also dominate when probability changes with scale. In the mean-one lognormal benchmark, $\tilde{R}_t^\tau = \exp(-v_t^2/2 + v_t\epsilon)$ with $\epsilon \sim N(0, 1)$, probability and severity satisfy

$$q_t^\tau = \frac{1 + a_t^\tau}{2} \quad m_t^\tau = \frac{2a_t^\tau}{1 + a_t^\tau} \quad \eta_q^\tau = \frac{a_t^\tau}{1 + a_t^\tau} \quad \eta_m^\tau = \frac{1}{1 + a_t^\tau} \quad (7)$$

At premiums of four percent or less, the severity elasticity exceeds 96 percent. Doubling the premium from two to four percent raises probability from 51 to 52 percent and severity from 3.92 to 7.69 percent. Severity accounts for 97.2 percent of that log premium increase.

These benchmarks explain why severity dominance alone cannot distinguish dispersion from distributional shape. They also give different probability restrictions—constant q under common scale and $q > 1/2$ under log-normality. A fixed nonlinear payoff map can produce still another pattern. The fixed-curvature example in Appendix A.3.2 shows that negative curvature can place q below one-half and make it decline with scale locally. The empirical exercise therefore recovers probability without imposing these families, then tests common scale using the wider quotation cross section.

3.3 RECOVERING THE COMPONENTS Recovering the decomposition requires one value of the risk-neutral distribution function, rather than its entire density. The first strike derivative of the put-price curve supplies that value (Breedon and Litzenberger, 1978).

Proposition 2 (Identifying probability and severity). *Fix a maturity τ and forward $F_{t,t+\tau}$. Under the maintained pricing assumptions, suppose there is no atom at the forward and the AtM-forward put has a positive price. Its price and local strike slope identify*

$$q_t^\tau = e^{r_t\tau} \left. \frac{\partial P_t(K, \tau)}{\partial K} \right|_{K=F_{t,t+\tau}} \quad m_t^\tau = \frac{a_t^\tau}{q_t^\tau} \quad (8)$$

Differentiating the discounted expected put payoff gives the discounted probability of finishing below the strike; the product identity then gives severity. Appendix A.3.3 provides the proof and the convexity bounds for discretely quoted strikes. The price–probability pair also has a familiar smile interpretation: at a fixed maturity, the gap $q_t^\tau - (1 + a_t^\tau)/2$ equals the local implied-volatility slope times a known positive factor. The decomposition expresses this local option information in probability and conditional-loss units (Appendix A.3.2).

For the baseline recovery, I retain the original quotation selection, parity-implied forwards, and selected maturities. At each maturity u , let $K_t^- < F_{t,t+u} < K_t^+$ be the nearest quoted put strikes bracketing the forward. Using bid–ask midpoints,

$$\hat{q}_t^u = e^{r_t u} \frac{P_t(K_t^+, u) - P_t(K_t^-, u)}{K_t^+ - K_t^-} \quad \hat{m}_t^u = \frac{a_t^u}{\hat{q}_t^u}$$

The put spread averages the cumulative probability across the bracket and approximates its value at the forward. It imposes neither lognormality nor fixed shape. Because the same bracket constructs the AtM premium, the midpoint interpolation is a recovery choice, rather than exact identification from finitely many quotations.

Carry-adjusted maturity weights preserve the original daily DPP. Monthly probability averages daily probabilities with carry weights; monthly severity additionally weights by shortfall probability. With these weights, the monthly adjusted premium satisfies

$$A_M \equiv \frac{\tilde{\chi}_M}{\tilde{D}_M} = q_M m_M. \quad (9)$$

These components summarize daily forward-looking distributions. They are not components of a realized calendar-month return, nor does maturity interpolation separately identify an exactly traded 30-day law. Appendix A.3.3

Table 2: Monthly Comovement and the Recession Premium

<i>Panel A: Monthly comovement</i>					
Component	Elasticity	HAC s.e.		95% interval	sd($\Delta \log z_M$)
Adjusted premium, A_M	—	—		—	0.172
Probability, q_M	−0.032	0.020		[−0.071, 0.006]	0.044
Severity, m_M	1.032	0.020		[0.994, 1.071]	0.183
<i>Panel B: The recession premium</i>					
Component	Expansion mean (%)	Recession mean (%)	$\mathcal{G}(z)$	HAC s.e.	Contribution (%) [95% interval]
Adjusted premium, A_M	1.77	3.17	0.576	0.109	100.0 8.5
Probability, q_M	42.31	44.41	0.049	0.014	[3.7, 13.3] 91.5
Severity, m_M	4.18	7.16	0.527	0.105	[86.7, 96.3]

Notes: January 1990–July 2026: 439 months, 438 adjacent changes, 36 recession months, and 403 expansion months. Recessions run from the month after an NBER peak through the trough. Phase means are arithmetic means; $\mathcal{G}(z)$ is the difference in mean log levels. Contributions equal $100\mathcal{G}(z)/\mathcal{G}(A)$; products of phase means need not equal the phase mean premium. Bartlett HAC uses six monthly lags and the correction $n/(n-2)$. Share intervals use the delta method and joint gap covariance. Intervals condition on recovered components.

gives the weighting formulas and sample checks. The baseline contains 8,791 daily observations and 439 months from January 1990 through July 2026.

3.4 SEVERITY DOMINANCE IN THE DATA

Monthly premium movements. The identity $\Delta \log A_M = \Delta \log q_M + \Delta \log m_M$ gives an exact covariance decomposition

$$\beta_q \equiv \frac{\text{Cov}(\Delta \log q_M, \Delta \log A_M)}{\text{Var}(\Delta \log A_M)} \quad \beta_m = 1 - \beta_q \quad (10)$$

These empirical elasticities are OLS slopes with an intercept on identical adjacent monthly observations. They measure comovement, rather than causal effects or local derivatives along a specified distributional family. Panel A of Table 2 reports a severity elasticity of 1.032, with a 95 percent interval of [0.994, 1.071]. The probability elasticity is −0.032, with an interval of [−0.071, 0.006]. Thus conditional severity moves approximately one-for-one with the premium, while probability contributes little to its monthly movements. Probability also fluctuates substantially less: its log-change standard deviation is 0.044, compared with 0.183 for severity. Figure 3 displays this contrast on a common proportional scale.

The recession premium. Let $\mathcal{G}(z)$ denote the recession mean log level minus its expansion counterpart. On identical observations,

$$\mathcal{G}(A) = \mathcal{G}(q) + \mathcal{G}(m) \quad (11)$$

Panel B shows that severity accounts for 91.5 percent of this log premium gap, with a 95 percent interval of [86.7, 96.3] percent. Mean severity rises from 4.18 percent in expansions to 7.16 percent in recessions, while mean probability rises from 42.31 to 44.41 percent. Probability makes a smaller but positive contribution: its log gap is 0.049, with an interval of [0.021, 0.077]. The result is severity dominance, rather than an exactly constant shortfall probability. The phase probabilities below one-half also differ from the lognormal benchmark, while the positive recession gap conflicts with a single common shape spanning the full sample.

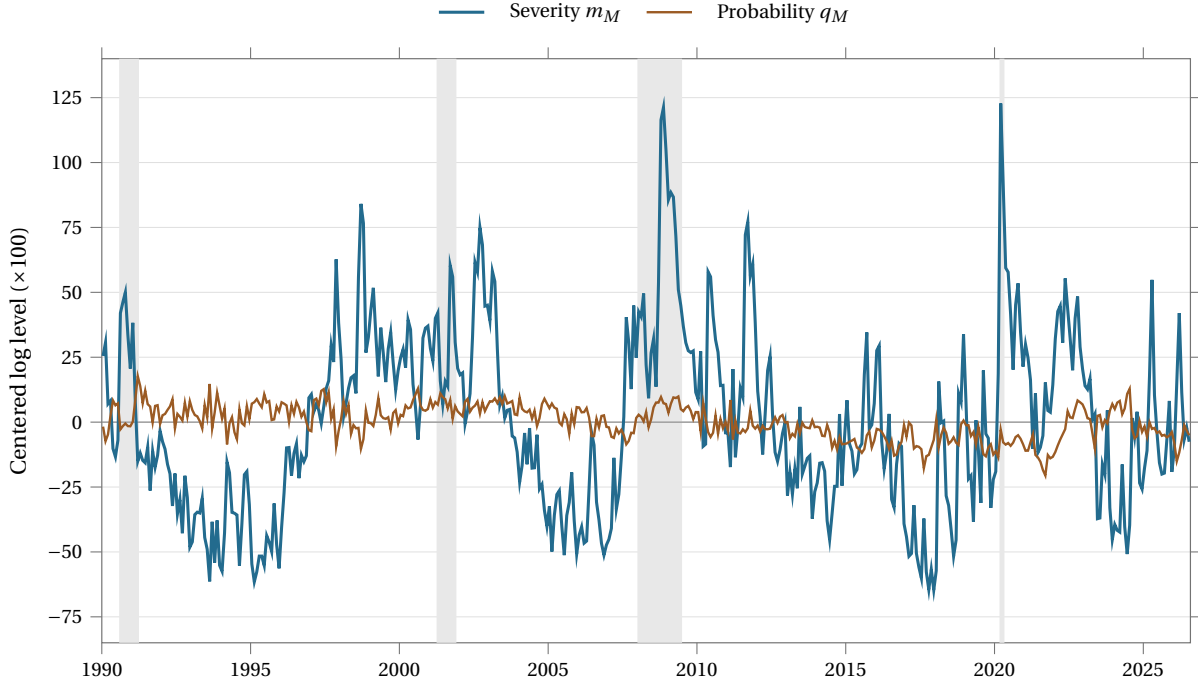


Figure 3: The DPP decomposition over time

Notes: Each series is $100(\log z_M - \overline{\log z})$ for $z \in \{q, m\}$, where the bar denotes the full-sample mean. The sum of the two components equals the centered log adjusted premium at every month; the premium is not separately plotted. Gray shading denotes recession months, from the month after the business-cycle peak through the trough. The sample runs from January 1990 through July 2026. The lines show recovered component estimates; uncertainty about component recovery is not represented by bands.

Recovery and stability. Table 3 compares five recoveries on common daily dates: put spreads, call spreads, a smooth curve, and entropy reconstructions with two reference distributions. Severity elasticities range from 1.016 to 1.055, and recession severity shares from 90.8 to 93.5 percent. The smooth and entropy methods jointly recover the premium and probability, so this agreement does not depend on fixing the original premium or on the baseline's local interpolation.

The conclusion also survives wider strike brackets, quotation-quality filters, minimum monthly coverage, rolling windows, alternative recession comparisons, and quarterly-horizon sampling. For example, omitting individual recessions yields severity shares of 90.0–93.3 percent, while quarterly severity loadings range from 0.951 to 1.023 across the reported recovery and sampling conventions. Appendices A.3.4 and A.3.5 report these checks, coverage, and conditional inference. The separate bid–ask quotation envelopes are too wide to establish dominance without recovery assumptions; the robust result is the agreement across the implemented recoveries, not identification of the decomposition from quotation intervals alone.

The evidence establishes a stable empirical composition of priced downside risk: conditional severity accounts for nearly all monthly premium comovement and roughly nine-tenths of the recession log premium. Figure 4 shows that the severity elasticity remains close to one across rolling ten-year windows, ranging from 0.987 to 1.097. The complementary probability elasticity remains close to zero, indicating that severity dominance is not confined to a single period. Because the DPP closely tracks the VIX, this decomposition gives economic content to the variation used in volatility-based macroeconomic studies. The remaining question is whether that composition can be explained by a common change in scale.

Table 3: Component Recovery on a Common Daily Sample

Method	Severity elasticity	Recession severity share (%)
	1.039	91.2
Baseline put spread	[0.997, 1.081]	[86.1, 96.3]
	1.055	92.0
Call spread	[1.017, 1.092]	[87.2, 96.9]
	1.042	91.9
Smooth curve	[1.009, 1.076]	[87.0, 96.9]
	1.016	90.8
Entropy, lognormal reference	[0.972, 1.060]	[86.0, 95.6]
	1.027	93.5
Entropy, uniform reference	[0.985, 1.069]	[89.1, 97.9]

Notes: All methods use the same 7,078 daily dates, 408 months, and 398 adjacent monthly changes. Brackets contain 95 percent intervals under the same six-lag HAC procedure. Smooth and entropy methods use their own jointly fitted premium. Inference conditions on recovered components.

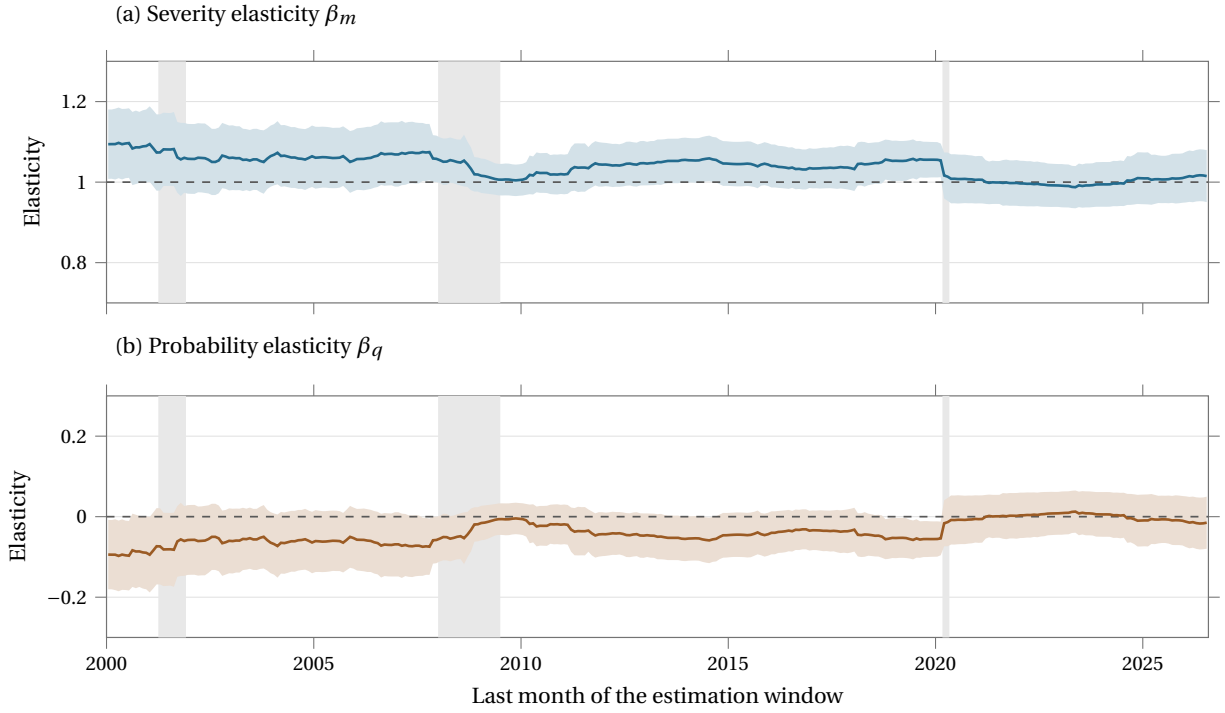


Figure 4: Probability and severity elasticities in rolling ten-year windows

Notes: Each estimate uses 120 adjacent monthly log changes and an intercept; the horizontal axis gives the final month of the estimation window. Bands are pointwise 95 percent confidence intervals using Newey–West covariance with six monthly lags and the correction $n/(n-2)$. The two elasticities sum to one in every window. The panels use equal vertical ranges, centered on the fixed-shape benchmarks of one for severity and zero for probability, shown by dashed lines. Gray shading denotes recession months at the window endpoint. Inference conditions on the recovered components; the bands describe uncertainty in the regression elasticities, not in monthly component levels.

3.5 BEYOND A COMMON CHANGE IN SCALE Severity dominance resembles the common-scale benchmark at the forward. That benchmark also restricts option prices away from the forward: the premium change determines how far the whole distribution expands. I test this additional restriction against quotations across dates and strikes, then relax it to distinguish probability changes from changes in the shape of conditional losses.

Write $X_t = \tilde{R}_t^\tau$ and $\mathcal{P}_t(k) = \mathbb{E}_t^Q[(k - X_t)^+]$, where $k = K/F$. Under Proposition 1,

$$\frac{\mathcal{P}_t(k)}{a_t} = C\left(\frac{k-1}{a_t}\right) \quad C(0) = 1 \quad (12)$$

Table 4: Compatibility of Scale Restrictions with Option Quotations

Restriction within each month	Months	Excluded	Outer fit	Unresolved
Full distribution: common scale	405	352	53	0
Conditional downside shape: constant q	405	345	55	5
Conditional downside shape: varying q	405	112	138	155

Notes: The sample covers 405 months between January 1990 and July 2026. All rows retain original quotation bands, fixed component anchors, and maturity construction. The last two rows allow the upside distribution to vary across dates. Exclusions are certified in an outer class; outer fits are verified numerical witnesses in that class. Unresolved outcomes establish neither exclusion nor a joint fit and remain in the denominators. Verified fits are carried into nested larger classes.

with one convex function C common to all dates in a comparison window. I allow a different candidate shape in each calendar month and ask whether any common shape within that month can fit the selected quotations inside their original bid–ask bands. The exercise retains the original forwards, carrying costs, component AtM midpoint anchors, and maturity weights. Each component law retains its quotation and moment restrictions; common scale applies to the carry-adjusted daily mixture. Thus the test conditions on the premium construction rather than optimizing its anchors within quotation intervals.

Table 4 reports the 6,966-date sample covering 405 months. Full common scale is excluded in 352 months, or 86.9 percent. Exclusions are certified in an enlarged distribution class that contains the strict positive-terminal-value benchmark. An exclusion therefore rules out that benchmark, whereas a verified fit in the outer class need not establish an attainable strictly positive distribution. These are joint quotation-compatibility results, rather than sampling rejection frequencies.

To isolate conditional downside shape, let

$$(1 - X_t) \mid \{X_t < 1\} = m_t U, \quad \mathbb{E}^Q[U] = 1, \quad m_t = \frac{a_t}{q_t}, \quad (13)$$

where U has a common conditional distribution within the month, while the upside distribution can vary freely across dates. For $k \leq 1$, this implies

$$\frac{\mathcal{P}_t(k)}{a_t} = H\left(\frac{(1-k)q_t}{a_t}\right), \quad H(z) = \mathbb{E}^Q[(U - z)^+]. \quad (14)$$

I compare an unknown common q with an unknown separate q_t at each date, retaining all component quotation and moment restrictions.

Most full-scale exclusions survive freeing the upside shape with 345 of the 352 excluded months remain excluded under constant- q conditional scale. Allowing probability to vary restores outer compatibility in 80 of those months. Nevertheless, 112 of all 405 months—27.7 percent—remain excluded even when each date has its own probability. In these months, probability adjustment alone cannot reconcile a common conditional downside shape with the maintained quotations. The varying- q calculation also yields 138 outer fits and 155 unresolved months, so 27.7 percent is a lower bound on the frequency of required conditional-shape departures under these restrictions.

For the full-scale exclusions, the median certified lower bound on the additional price-band allowance is 1.39 percent of the daily adjusted premium. This is a minimum price discrepancy beyond the original bid–ask bands, not an estimate of a macroeconomic effect or a sufficient allowance to restore compatibility. Appendix A.3.6 provides the construction, numerical criteria, price-distance diagnostics, and window comparisons.

The section therefore supplies two complementary empirical restrictions. At the forward, movements in the

price of protection are overwhelmingly associated with conditional shortfall severity. Across strikes, those movements frequently require more than a common rescaling, including changes in conditional downside shape in a substantial subset of months. Together, these results sharpen what a model matching stock-market volatility must explain: the composition of expected shortfall and its relation to the wider option-price cross section. The next section examines their macroeconomic interpretation.

4 INTERPRETING UNCERTAINTY SHOCKS

Section 3 establishes that conditional severity accounts for most movements in the price of downside protection. Does this common composition also imply a common macroeconomic response? I address this question using the realized-volatility and future-volatility-news innovations of Berger et al. (2020). Both raise the DPP primarily through conditional severity, yet their employment responses remain different at the point estimates after equalizing the impact increase in severity. I then explain how these observations restrict a structural model's mapping from economic risk to option prices and macroeconomic activity.

4.1 SEVERITY DOMINANCE AND MACROECONOMIC RESPONSES Berger et al. (2020) distinguish an innovation in realized stock-market volatility from news about future volatility orthogonal to the contemporaneous realized-volatility innovation. They find a larger employment decline following the realized-volatility innovation, despite comparable cumulative effects on expected future volatility. This distinction connects uncertainty to the macroeconomic literature on news: information about future conditions can affect current decisions before those conditions materialize (Beaudry and Portier, 2006; Barsky and Sims, 2011; Beaudry and Portier, 2014). The DPP decomposition adds two questions: whether the innovations reprice different shortfall components, and whether their employment contrast survives expressing them in common impact-severity units.

Innovation construction and option responses. I reproduce the authors' innovation construction using their February 1983–December 2014 sample. Following their replication code, the realized-volatility innovation is scaled to match the news innovation's cumulative effect on expected future volatility over months 1–23. On the overlapping 1990–2014 option sample, I estimate local projections of $\log z_{t+h} - \log z_{t-1}$ on both innovations and lagged controls, for $z \in \{A, q, m\}$. The monthly option measures summarize the final three trading sessions, rather than the full-month averages in Section 3, and preserve $A = qm$. Identical observations and regressors therefore imply

$$\text{IRF}_{\log A}(h) = \text{IRF}_{\log q}(h) + \text{IRF}_{\log m}(h) \quad (15)$$

The identity holds in every joint bootstrap draw. Macroeconomic projections use the same observations, innovations, and controls, and inference re-estimates the first stage and volatility normalization in each draw. Appendix A.2.5 gives the estimation and sampling details.

Table 5 shows that both impact premium responses are dominated by severity. The probability-to-premium response ratios are -0.031 for realized volatility and -0.082 for news. The positive severity responses rule out an impact premium increase operating entirely through shortfall probability under this identification and recovery. The imprecise ratio difference does not establish equal compositions, and the probability intervals do not establish an exactly zero response. These innovation-specific ratios are distinct from the aggregate monthly covariance loading in Section 3.

Table 5: Impact responses of the protection premium and its components

<i>A. Cumulative log responses, multiplied by 100</i>			
Innovation	Premium A	Probability q	Severity m
Realized volatility	8.76 [3.87, 17.42]	-0.27 [-1.54, 0.52]	9.03 [3.93, 18.56]
Future-volatility news	11.44 [10.26, 12.14]	-0.94 [-2.03, 0.22]	12.38 [10.84, 13.33]
<i>B. Probability-to-premium impact response ratios</i>			
	Realized volatility	News	RV minus news
Ratio or difference	-0.031 [-0.120, 0.060]	-0.082 [-0.187, 0.019]	0.051 [-0.093, 0.202]

Notes: There are 291 impact origins in 1990–2014. Brackets report 95% pointwise percentile intervals from 999 joint circular-block bootstrap draws with 24-month blocks. The first-stage VAR, innovations, and realized-volatility normalization are re-estimated in each draw. Ratios and their difference are formed within each draw. Intervals condition on the recovered daily option components; recovery methods are compared separately in Appendix A.2.5.

The employment contrast. Table 6 reports the twelve-month activity responses under the equal-future-volatility normalization. Manufacturing employment falls by 0.477 after realized volatility, while its estimated response to news is 0.151, in units of 100 times cumulative log changes. The direct contrast is -0.628 , with a 95 percent simultaneous interval of $[-1.159, -0.097]$ covering both activity contrast paths over horizons 0–24. Industrial production has a negative pointwise contrast, but its simultaneous interval includes zero. Thus employment supplies the stronger evidence of a difference, reproducing the qualitative pattern of Berger et al. (2020) on the option sample. The individual news interval includes modest contractions, so the comparison does not establish that news has no contractionary effect.

Table 6: Twelve-month macroeconomic responses

		RV minus news	
	RV	News	
			Pointwise Simultaneous
<i>Manufacturing employment</i>			
Estimate	-0.477	0.151	-0.628 -0.628
95% interval	[-0.758, -0.117]	[-0.056, 0.362]	[-0.929, -0.224] [-1.159, -0.097]
<i>Industrial production</i>			
Estimate	-0.762	0.421	-1.183 -1.183
95% interval	[-1.428, -0.081]	[-0.121, 0.998]	[-1.966, -0.242] [-2.539, 0.173]

Notes: Entries are 100 times cumulative log responses at $h = 12$ (275 origins). Relative to the 291 impact origins, twelve origins in 2014 are lost because $t + 12$ exceeds December 2014, and four earlier origins are lost because option components are missing at $t + 12$. Every column uses normalization re-estimated in each bootstrap draw. Individual-innovation intervals are pointwise. The final column uses a simultaneous band for both RV-minus-news contrast paths over horizons 0–24. Contrasts are formed within joint draws, rather than by subtracting interval endpoints.

Comparing responses at equal impact severity. Equal cumulative effects on future volatility need not produce equal impact severity responses. Figure 5 therefore scales each innovation to raise $\log m$ by 0.01 on impact. Let $e_j(h)$ denote the cumulative employment log response and $b_j = \text{IRF}_{\log m, j}(0)$ impact severity, for $j \in \{R, N\}$. A common proportional mapping from impact severity to employment requires

$$\frac{e_R(h)}{b_R} = \frac{e_N(h)}{b_N} \iff e_R(h)b_N - e_N(h)b_R = 0 \quad (16)$$

for nonzero impact severities. At twelve months, the normalized employment contrast is -0.065 in units of 100 times log changes per 0.01 impact log-severity increase. Its pointwise interval is $[-0.098, -0.020]$, and its simultaneous interval is $[-0.124, -0.006]$ over both activity contrasts at horizons 0–24.

The cross-product in (16) checks the same economic restriction without a random denominator. Using the fixed point-estimate product to express each draw in comparable units gives a pointwise interval of $[-0.104, -0.023]$ and a simultaneous interval of $[-0.1305, 0.0005]$. Both formulations therefore provide pointwise evidence *against* the common proportional mapping, but their simultaneous conclusions differ: the ratio interval excludes zero, while the cross-product interval narrowly includes it. The twelve-month comparison was among the reporting horizons selected before computing the macroeconomic estimates, after examining the component paths, rather than a sole confirmatory test. Appendix A.2.5 reports the joint inference and reporting protocol.

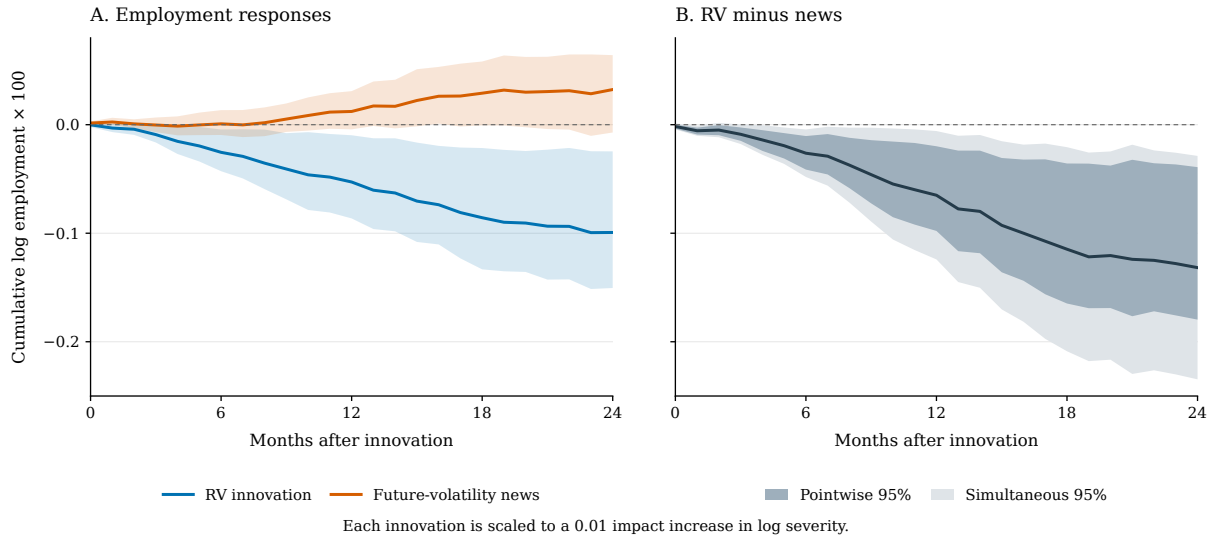


Figure 5: Manufacturing employment responses at equal impact severity.

Notes: Each innovation is scaled to increase $\log m$ by 0.01 on impact. Employment responses are 100 times cumulative log changes. Colored shading in the left panel and darker shading in the right panel give 95% pointwise intervals. Lighter shading in the right panel is the ratio-based 95% simultaneous band covering both employment and industrial-production contrasts over horizons 0–24, although only employment is plotted. The twelve-month employment contrast is -0.065 in units of 100 times log changes, with a 95% simultaneous interval of $[-0.124, -0.006]$. Each of the 999 joint draws uses its own impact severity response as the denominator. Under each draw's re-estimated volatility normalization, the smallest impact log-severity denominators are 0.01057 for realized volatility and 0.09057 for news. The cross-product check is reported separately in the text. Later severity paths and other variables are not held fixed. Sample: 1990–2014.

This normalization compares responses in common impact units; it does not hold subsequent severity paths or other equilibrium responses fixed. It consequently evaluates the proportional restriction in (16), rather than identifying a causal effect of severity or testing whether the full priced-distribution path explains employment. An exploratory equal-impact-premium normalization gives a similar twelve-month contrast of -0.068 , with a ratio-based simultaneous interval of $[-0.127, -0.008]$. An unrestricted mapping from both impact components is not tested: with two innovations, two freely chosen component weights can fit the two employment responses. The appendix reports that calculation and the direct comparison of subsequent severity paths.

Call-spread and smooth-curve intersections preserve severity dominance and the negative twelve-month employment contrast at equal impact severity under the ratio-based simultaneous bands. The entropy intersections retain similar employment point estimates but have substantially less coverage, and their bands include zero. Both innovations also increase conditional priced gain magnitudes, so the finding does not describe an exclusively downside movement. Appendix A.2.5 reports the full recovery comparisons and upside responses.

The decomposition therefore adds a substantive restriction to the macroeconomic comparison: both innovations increase the price of protection primarily through conditional shortfall severity, yet their estimated employment responses differ even in common impact severity units. The evidence against a common proportional response has pointwise support under both inference formulations, with stronger simultaneous support under the ratio formulation. Severity dominance identifies a shared dimension of repricing, but that dimension alone does not give the two innovations a common employment response.

4.2 WHAT A STRUCTURAL MODEL MUST EXPLAIN The option restrictions concern the priced distribution generated by a model's equilibrium equity payoffs and stochastic discount factor. To make this mapping explicit, fix the horizon and write $X_t = S_{t+\tau}/F_{t,t+\tau}$ and $L_t = (1 - X_t)^+$. For the stochastic discount factor $M_{t,t+\tau} > 0$, define

$$w_t = \frac{M_{t,t+\tau}}{\mathbb{E}_t^{\mathbb{P}}[M_{t,t+\tau}]} \quad F_{t,t+\tau} = \mathbb{E}_t^{\mathbb{P}}[w_t S_{t+\tau}]$$

Under the maintained deterministic carrying-cost assumptions, w_t maps physical probabilities into the pricing probabilities used above.⁶ For the shortfall event $\mathcal{D}_t = \{X_t < 1\}$, let $p_t = \mathbb{P}_t(\mathcal{D}_t)$, $\kappa_t = \mathbb{E}_t^{\mathbb{P}}[w_t | \mathcal{D}_t]$, and $m_t^P = \mathbb{E}_t^{\mathbb{P}}[L_t | \mathcal{D}_t]$. For $p_t > 0$, the priced components satisfy

$$q_t = p_t \kappa_t \quad m_t = m_t^P + \frac{\text{Cov}_t^{\mathbb{P}}(w_t, L_t | \mathcal{D}_t)}{\kappa_t}$$

Probability combines physical frequency with average valuation in shortfall states; severity combines physical conditional losses with their relative valuation within those states. The forward threshold is itself endogenous, so a valuation change can also change the measured physical components. Appendix A.2.4 derives the identities and examines restricted valuation experiments.

The empirical results restrict this joint mapping without identifying its physical and valuation terms separately. Near-zero aggregate probability comovement does not require constant p_t or κ_t , or a zero probability response to every shock. Likewise, a primitive disaster-probability shock need not appear mainly in q_t : by placing more weight on very large losses, it can increase severity within the broader event of any shortfall. The empirical components therefore discipline a model's observables, rather than assigning names to its primitive shocks.

A structural interpretation must account for two linked sets of observations. Section 3 establishes the composition of premium movements, probability levels, and departures from common scale across strikes. The present section establishes severity-dominated impact repricing for two identified volatility innovations with different estimated employment responses, including after impact-severity normalization. Matching implied volatility or a near-unit severity loading alone does not match this joint evidence.

For a model disciplined by implied stock-market volatility, such as Basu and Bundick (2017), the corresponding comparison requires pricing the model's equity protection claims and applying the empirical identification to simulated observables at comparable horizons. Berger et al. (2020) already apply their identification to simulations of that model. The additional question is whether its equilibrium option prices reproduce the composition and quotation restrictions in Section 3 alongside the component and employment responses studied here. These results give that comparison observable content; they do not establish that a particular model fails it. Severity dominance sharpens the interpretation of movements in the uncertainty proxy, while their macroeconomic consequences depend on the economic innovations and equilibrium propagation that generate them.

⁶Normalizing by the zero-coupon bond price generally defines the horizon-specific forward pricing measure. It coincides with the risk-neutral measure used here under the maintained assumptions. Payoffs and the discount factor must use the same units.

5 CONCLUSION

The normalized at-the-money forward put tracks the VIX, preserves the principal macroeconomic responses in the specifications examined, and separates the priced probability of a shortfall from its conditional severity. Conditional severity accounts for almost all monthly premium comovement and most of the recession premium, and conditional gains rise with it. Because priced shortfall probabilities average below one-half, conditional losses remain larger than gains.

These regularities do not reduce to a common change in scale. For the recovered series, the probability level and its recession increase rule out exact lognormality, a single shape spanning the sample, and a constant local probability gap. Conditional on the maintained forward and premium anchors, quotations across strikes exclude common scale within 352 of 405 covered months and require a change in conditional downside shape in at least 112. Realized-volatility and future-volatility-news innovations both raise protection costs mainly through severity, yet are followed by different estimated employment responses at equal impact severity.

A model whose uncertainty process is disciplined by implied volatility has matched one margin of option prices. Its equilibrium quotations, constructed for the same payoff and horizon and passed through the same component recovery and innovation identification, provide a natural test of the remaining restrictions.

DECLARATION OF ARTIFICIAL INTELLIGENCE USAGE

The first version of this paper was drafted without any AI assistance. I used ChatGPT-6.1 to help produce the replication package available at XX. A draft of the manuscript was analyzed by ChatGPT-6.0 and suggestions were made to improve the readability of the paper. I implemented a subset of these recommendations and take full responsibility for the contents of this article.

DATA AVAILABILITY STATEMENT

A replication package for results in the paper and Appendices A1 and A2 is available at doi: 10.5281/zenodo.23106238.

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A SUPPLEMENTARY APPENDIX (ONLINE APPENDIX)

This appendix documents the data and construction in Appendix A.1, the macroeconomic comparisons and physical-to-priced mapping in Appendix A.2, and the derivations and robustness of the DPP decomposition in Appendix A.3. The main estimates and their supporting robustness comparisons are documented together.

A.1 DATA AND DPP CONSTRUCTION

A.1.1 S&P 500 INDEX OPTIONS The options data are daily end-of-day S&P 500 Index (SPX) puts and calls on a wide range of strikes obtained from DeltaNeutral (HistoricalOptionData.com) from January 1990 through July 2026. The data include traditional monthly SPX contracts (morning settlement) and SPXW contracts (afternoon settlement). This distinction matters for measuring time to maturity and matching put and call contracts as an identical strike and calendar expiration date need not imply identical settlement terms. For each contract, the data report its full option identifier, put or call type, strike, expiration date, bid and ask quotations, last transaction price, trading volume, and open interest, together with the underlying index level at the recorded snapshot. Option prices and strikes are measured in index points, and the contract multiplier is \$100 per index point. Dividing a put price by the underlying index level therefore expresses its premium per dollar of index exposure. Trading volume and open interest are measured in numbers of contracts.

The distinction between quoted and traded prices is relevant for assessing the quality of these observations. The bid and ask describe the market quotation at the daily snapshot, whereas the last transaction price may refer to a trade on an earlier date. An option can therefore have an end-of-day quotation even when its reported daily volume is zero.

A.1.2 QUOTATION SELECTION AND PREMIUM COVERAGE The baseline uses bid–ask midpoints, requiring a non-negative bid, a strictly positive ask, and an ask no lower than the bid. No minimum volume or maximum relative spread is imposed. Puts and calls used in parity calculations are matched by snapshot, strike, expiration, contract family, and settlement terms. For each selected maturity, the forward is the median parity-implied value from up to three matched strikes closest to the contemporaneous index. The daily three-month Treasury bill rate is converted to continuously compounded units; missing rates are carried forward for at most three calendar days. Prices are interpolated first across the nearest strict put strikes bracketing that maturity’s forward, then across the eligible maturities bracketing 30/365 years. The total strike-bracket width cannot exceed five percent of the forward. There is no additional limit on the distance between bracketing maturities. Neither strike nor maturity extrapolation is used.

The premium is available on 8,825 of 9,212 regular trading sessions from January 2, 1990 through July 31, 2026. Of the 387 missing sessions, two have no source SPX quotations, 379 lack a usable maturity bracket, and six fail the strike-width restriction. Of the maturity failures, 359 occur between the 14th and 21st of the month, when the next standard monthly expiration is more than 30 days away; the remaining 20 occur between September 18 and October 15, 1992. No missing date is filled. Arithmetic averages of available daily premiums yield 439 monthly observations. Component recovery retains 8,791 dates; its exclusions and matched-sample effects are documented in Appendix A.3.5.

A.1.3 TRADING VOLUME AND COVERAGE Volume measures trading during the day; open interest measures outstanding positions and is updated separately, before the trading session. Figure A.1 summarizes recorded SPX trading activity over the available sample. The volume calculations use observations with positive reported volume,

after removing repeated records for the same contract and date. Panel (a) reports 21-weekday moving averages of daily put and call volume, summed across strikes and expirations. Trading activity increases substantially during the 2000s, and put volume generally exceeds call volume.

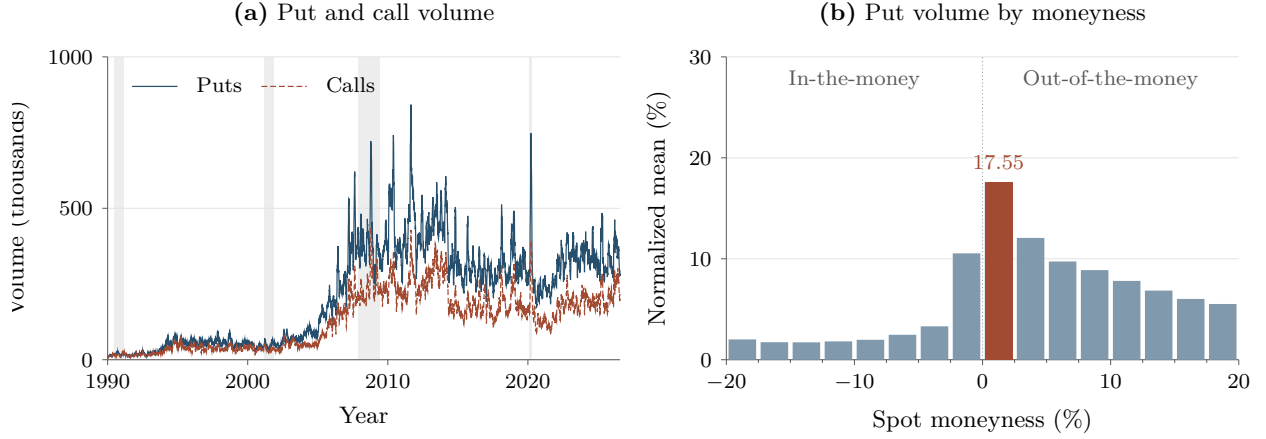


Figure A.1: SPX trading activity. Panel (a) shows 21-weekday moving averages of aggregate daily volume, in thousands of contracts. Weekdays without recorded trading enter the moving average as zeros; shaded intervals mark NBER business-cycle peaks to troughs. Panel (b) shows mean volume per positive-volume put contract-date in 2.5-percentage-point spot-moneyness bins, normalized to sum to 100 over the displayed range, $[-20, 20]$ percent. Positive moneyness denotes out-of-the-money puts. Both panels pool expirations. Source: DeltaNeutral option data; NBER business-cycle chronology.

Panel (b) describes trading activity across strikes using spot moneyness, $m_{it} = 100(S_t - K_i)/S_t$, where S_t is the underlying index level and K_i is the strike. This differs from the paper's at-the-money put, which is defined relative to the forward price. For each moneyness bin, I compute mean volume per active put contract-date and normalize these means to sum to 100 across bins. Average activity is greatest in the 0–2.5 percent out-of-the-money bin, whose normalized mean is 17.75 percent, and is generally lower farther from the money.

A.2 MACROECONOMIC REPLICATION AND STRUCTURAL INTERPRETATION This appendix examines whether replacing the VIX/VXO with the thirty-day Downside Protection Premium (DPP) changes the identification of uncertainty shocks or their estimated macroeconomic effects. Within each proxy comparison, I replace only the uncertainty proxy; the sample, macroeconomic controls, lag order, deterministic terms, and recursive ordering are all held fixed. These comparisons assess the preservation of the principal macroeconomic responses, rather than exact agreement in event dates or inference classifications. The remaining material derives the physical-to-priced mapping and documents the volatility-innovation comparisons in Section 4. Derivations and robustness supporting the DPP decomposition, including the quotation-based scale restrictions, appear in Appendix A.3.

A.2.1 BLOOM-STYLE EVENT IDENTIFICATION Following the threshold approach of Bloom (2009), I apply the HP filter ($\lambda = 129,600$) to monthly DPP and VIX levels over January 1990–July 2026. Both series are averaged over identical daily dates. I standardize each cyclical component by its sample standard deviation and identify the peak month within each contiguous episode exceeding 1.65 standard deviations consistent with Bloom (2009).

Figure A.2 shows closely aligned cycles, with correlations of 0.9926 in levels and 0.9885 in logarithms. At the 1.65 cutoff, each level-based series identifies ten events, eight with identical peak months. The VIX additionally identifies October 1990 and November 1997, while the DPP additionally identifies May and October 2022. October

1990 narrowly misses the DPP cutoff, with a standardized cycle of 1.6499 compared with 1.6539 for the VIX. The logarithmic specification identifies fourteen events for each proxy, eleven with identical peak months.

The largest events coincide. At a common cutoff of 2.02 standard deviations, both level-based series identify the same six peak months: September 1998, September 2002, November 2008, September 2011, March 2020, and April 2025. The largest events therefore coincide, although the baseline threshold selects some different episodes. The October 1990 discrepancy is particularly sensitive to the cutoff; the other differences are not merely rounding errors.

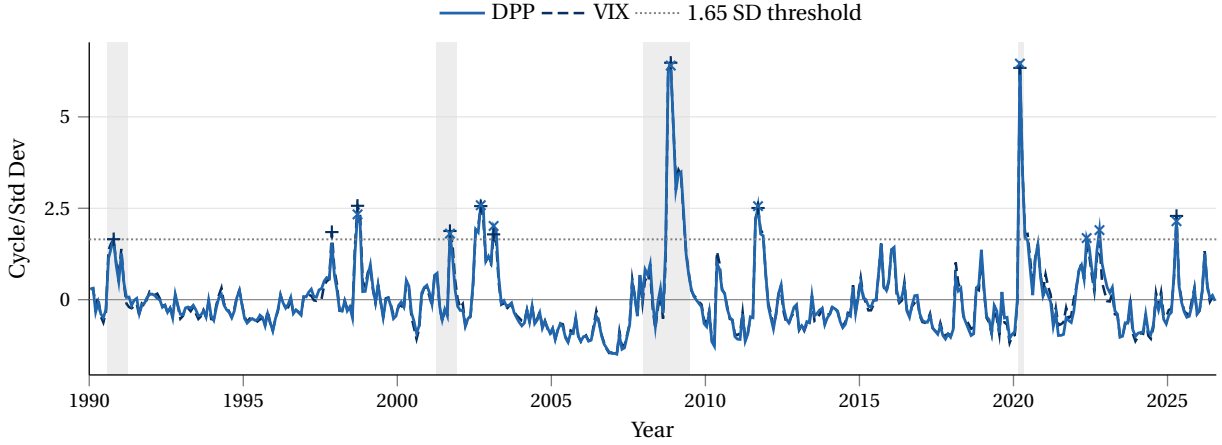


Figure A.2: Bloom-style uncertainty events, January 1990–July 2026. Standardized HP cycles of monthly DPP (solid) and VIX (dashed), using levels in Panel (a) and logarithms in Panel (b), with $\lambda = 129,600$. The dotted line marks 1.65 standard deviations; symbols identify peak months within contiguous above-threshold episodes. Grey shading denotes NBER recessions, from the month after the peak through the trough.

A.2.2 MONTHLY VAR OF LEDUC AND LIU (2016) The monthly comparison follows the recursive identification of Leduc and Liu (2016), using a frequentist log-proxy VAR, as opposed to their Bayesian approach. For uncertainty proxy $j \in \{\text{DPP}, \text{VIX}\}$, define

$$X_m^{(j)} = \begin{pmatrix} \log \bar{j}_m \\ U_m \\ \pi_m \\ i_m \end{pmatrix} \quad X_m^{(j)} = a^{(j)} + \sum_{\ell=1}^p A_\ell^{(j)} X_{m-\ell}^{(j)} + u_m^{(j)}$$

Here $\bar{j}_m$ is the matched monthly average of the chosen uncertainty proxy, U_m is the unemployment rate, i_m is the three-month Treasury bill rate, and $\pi_m = 100 [\log(\text{CPI}_m) - \log(\text{CPI}_{m-12})]$. Data are the FRED series UNRATE, CPIAUCSL, and TB3MS. Following Leduc and Liu (2016), the baseline specification includes a constant and twelve lags, and identification uses a lower-triangular Cholesky factor of the residual covariance matrix, with uncertainty ordered first.

I report two shock normalizations. The first uses each model's own one-standard-deviation uncertainty innovation. The second scales both responses to an initial increase of 0.1 log unit in the uncertainty proxy, corresponding to a 10.52 percent proportional increase. This comparison separates differences in propagation from differences in the estimated innovation scale. Uncertainty bands use 1,000 paired residual-bootstrap replications. The standardized uncertainty innovations have a correlation of 0.9836.

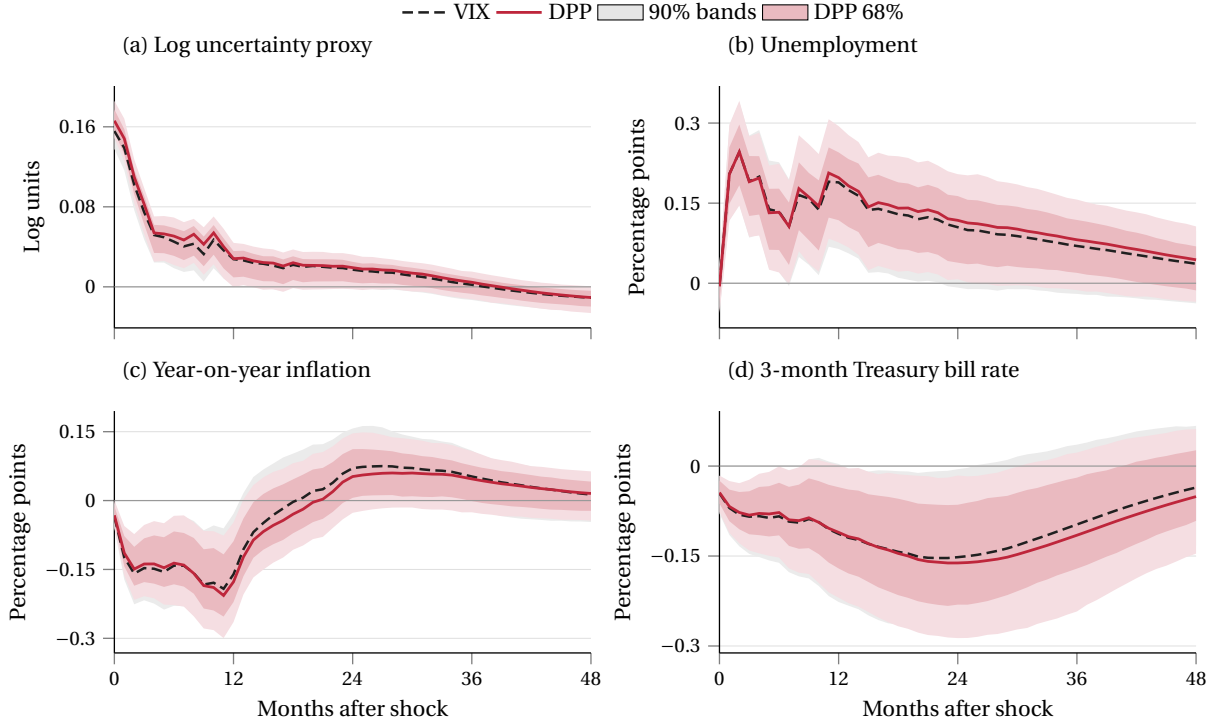


Figure A.3: Monthly responses to uncertainty innovations measured using the DPP and VIX. The solid line denotes the DPP specification and the dashed line the VIX specification. Each shock is normalized to its own one-standard-deviation innovation. The specifications use January 1990–September 2025, twelve lags, a constant, and uncertainty-first recursive identification. The shaded regions show pointwise 90 percent bootstrap intervals for both specifications and a darker 68 percent interval for the DPP. The uncertainty response is in log units; the remaining responses are in percentage points. Intervals use 1,000 paired residual-bootstrap replications.

Figure A.3 shows closely aligned impulse responses, with some differences in the inflation response. For example, under the own-standard-deviation normalization, the inflation-response difference is negative with its 90 percent interval excluding zero at months sixteen through twenty. The corresponding unemployment and interest-rate difference intervals contain zero throughout the 48 months.

As a robustness, I consider a common AIC-selected lag order and differing sample endpoints. Table A.1 shows that the monthly innovation correlation remains above 0.981 across these specifications. The close correspondence therefore survives changing the lag order, using an endpoint close to the Leduc–Liu sample, and excluding the pandemic period.

A.2.3 QUARTERLY VAR OF BASU AND BUNDICK (2017) The quarterly exercise uses the eight-variable specification of Basu and Bundick (2017). The variables are ordered as follows: log uncertainty, log real output per capita, log real consumption per capita, log real investment per capita, log hours per capita, log price level, log M2, and the shadow policy rate. The original macroeconomic observations and transformations are retained in every comparison. Each specification includes a constant and four lags, with uncertainty ordered first. The common sample, 1990Q1–2014Q4, contains 100 quarters and 96 fitted observations. I first reproduce the authors’ VIX point estimates. The maximum absolute discrepancy across the stored impulse responses is below 10^{-10} . I then replace the uncertainty series with the DPP while holding the seven macroeconomic controls and all other specification choices fixed. The comparisons include the authors’ original VIX, a matched VIX series, and the authors’ original

Table A.1: Correlations between Identified DPP and Implied-Volatility Innovations

Specification and benchmark	Lags	Observations	Correlation
Monthly: VIX, baseline	12	417	0.9836
Monthly: VIX, common AIC order	14	415	0.9834
Monthly: VIX, through October 2013	12	274	0.9839
Monthly: VIX, through December 2019	12	348	0.9814
Quarterly: matched VIX	4	96	0.9845
Quarterly: author VIX	4	96	0.9775
Quarterly: author VXO	4	96	0.9867

Note: Entries are Pearson correlations between standardized recursively identified uncertainty innovations. Observation counts refer to fitted observations after lags. The baseline monthly sample retains the September 2025 endpoint of the saved macroeconomic replication; extending the option sample through July 2026 does not re-estimate this VAR beyond that endpoint. All quarterly comparisons use 1990Q1–2014Q4. Matched VIX uses the same daily observations and monthly aggregation as the DPP; author VIX and VXO retain the published quarterly series.

VXO. For the matched VIX and DPP, quarterly levels are equal-weighted averages of the three monthly arithmetic means, followed by the logarithmic transformation. Uncertainty band construction follows the authors' simulation conventions; however, I use 1,000 posterior draws per model, compared with 5,000 in the original simulation.

The DPP innovations correlate 0.9845 with the matched VIX innovations, 0.9775 with the authors' original VIX innovations, and 0.9867 with the authors' VXO innovations. Figure A.4 shows impulse responses between the DPP and VIX are nearly identical. At each horizon, I classify a macroeconomic response according to whether its interval is entirely positive, entirely negative, or contains zero. At the 90 percent level, agreement is 140 of 140 for VIX and 136 of 140 for VXO. Neither comparison switches a response from positive to negative, although the VXO comparison differs in whether four intervals include zero. The three uncertainty proxies produce nearly identical real-activity troughs: output falls by 0.167–0.182 percent and consumption by 0.149–0.151 percent in quarter three, while investment falls by 0.445–0.464 percent and hours by 0.227–0.244 percent in quarter four. These responses are measured as 100 times the log deviation following each model's own one-standard-deviation uncertainty innovation, with the sample, macroeconomic controls, lag order, and recursive identification held fixed.

Taken together, the exercises show that the principal macroeconomic conclusions are robust to replacing the implied-volatility proxy with the DPP in the specifications examined here.

A.2.4 PHYSICAL RISK, VALUATION, AND BENCHMARK RESTRICTIONS

Mapping physical risk into the priced components. For the normalized kernel, $\mathbb{E}_t^{\mathbb{P}}[w_t] = 1$ and $\mathbb{E}_t^{\mathbb{Q}}[f(X_t)] = \mathbb{E}_t^{\mathbb{P}}[w_t f(X_t)]$ for integrable payoffs. With $D_t = \{X_t < 1\}$ and $L_t = (1 - X_t)^+$, define

$$p_t = \mathbb{P}_t(D_t) \quad \kappa_t = \mathbb{E}_t^{\mathbb{P}}[w_t | D_t] \quad m_t^{\mathbb{P}} = \mathbb{E}_t^{\mathbb{P}}[L_t | D_t] \quad (\text{A.1})$$

Assuming $p_t > 0$, conditioning on the shortfall event gives

$$q_t = p_t \mathbb{E}_t^{\mathbb{P}}[w_t | D_t] = p_t \kappa_t \quad a_t = p_t \mathbb{E}_t^{\mathbb{P}}[w_t L_t | D_t]$$

Dividing the second expression by the first and expanding the conditional product expectation yields

$$m_t = m_t^{\mathbb{P}} + \frac{\text{Cov}_t^{\mathbb{P}}(w_t, L_t | D_t)}{\mathbb{E}_t^{\mathbb{P}}[w_t | D_t]} \quad (\text{A.2})$$

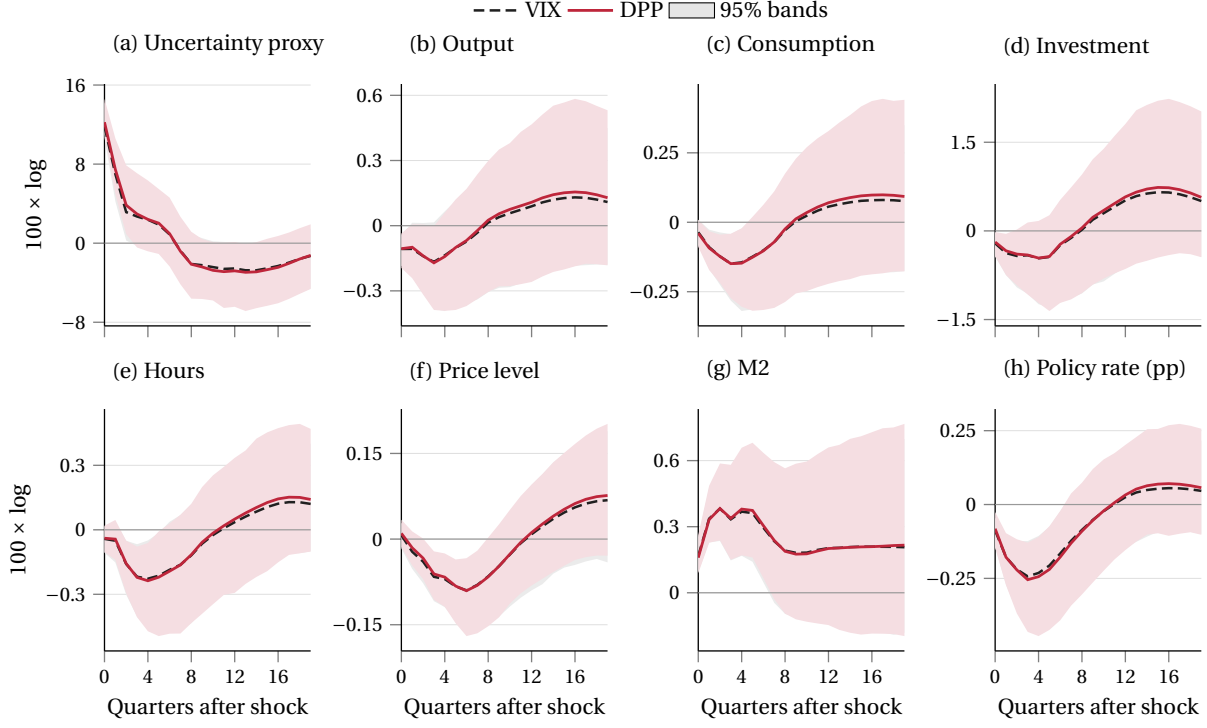


Figure A.4: Quarterly DPP and VIX responses in the Basu–Bundick specification. The solid line denotes the DPP and the dashed line the matched VIX specification. Both use the same original macroeconomic controls, 1990Q1–2014Q4, four lags, a constant, and uncertainty-first recursive identification. Shading denotes modelwise pointwise 95 percent posterior intervals based on 1,000 draws. Shocks are normalized to each model’s own one-standard-deviation uncertainty innovation. Responses of logarithmic variables are multiplied by 100; the policy-rate response is in percentage points. The direct VXO comparison is reported separately in the text and numerical replication outputs.

The probability identity also gives $\Delta \log q_t = \Delta \log p_t + \Delta \log \kappa_t$. A constant priced probability need not imply a constant physical probability or a constant average downside valuation. Any offset between them must follow from the model’s equilibrium; the kernel cannot be chosen afterward to reconcile an otherwise unrestricted physical distribution with the option evidence.

All quantities refer to the same forward-normalized payoff. When the forward changes, the event D_t changes with it. A change in p_t therefore need not have the same sign as a change in the probability of a fixed loss relative to the current equity price or of an adverse macroeconomic outcome.

Variation entirely through priced shortfall frequency. If conditional priced severity remains fixed at $\bar{m}$, then

$$a_t = q_t \bar{m} \implies \Delta \log a_t = \Delta \log q_t \quad \Delta \log m_t = 0 \quad (\text{A.3})$$

This composition is consistent with positive terminal equity and the forward normalization. For example, assign probability $\theta_t \in (0, 1)$ to $X_t = 1 - \ell$, with $0 < \ell < 1$, and probability $1 - \theta_t$ to $X_t = 1 + \ell \theta_t / (1 - \theta_t)$. The mean is one, $q_t = \theta_t$, and $m_t = \ell$. If this is the sole source of premium variation, the probability loading is one and the severity loading is zero. This example restricts the equilibrium distribution. A primitive disaster-frequency shock need not have this composition, because it can also change equity payoffs, valuation, and the forward threshold.

Restricted changes in valuation. A restricted valuation experiment clarifies the content of the severity identity. Hold the physical payoff distribution and the forward fixed, and perturb the normalized kernel from w to $w + \epsilon h$, preserving positivity. Normalization and the fixed forward require $\mathbb{E}^{\mathbb{P}}[h] = \mathbb{E}^{\mathbb{P}}[hX] = 0$. To raise the premium while leaving shortfall probability unchanged, the perturbation must also satisfy

$$\mathbb{E}^{\mathbb{P}}[h\mathbf{1}_D] = 0 \quad \mathbb{E}^{\mathbb{P}}[hL] > 0 \quad (\text{A.4})$$

Valuation must increase the priced loss without increasing total priced probability on the shortfall event. A uniform increase in the normalized kernel throughout D cannot do this in the fixed-payoff experiment. These are weighted-average conditions. They do not require pointwise monotone reweighting, and the general identities must be used when payoffs, physical probabilities, or the forward also move.

One useful level comparison is available under an additional restriction. If the projected kernel $g_t(x) = \mathbb{E}_t^{\mathbb{P}}[w_t | X_t = x]$ is nonincreasing below the forward, worse losses receive weakly greater weight and $\text{Cov}_t^{\mathbb{P}}(w_t, L_t | D_t) \geq 0$. Then $m_t \geq m_t^{\mathbb{P}}$. This condition must be checked in the model. It neither follows from the decomposition nor implies that the valuation contribution rises after a shock.

Joint option-price objects. The premium, shortfall probability, and idealized log-contract volatility summarize different features of the same priced distribution

$$a_t = \mathbb{E}_t^{\mathbb{Q}}[(1 - X_t)^+] \quad q_t = \mathbb{Q}_t(X_t < 1) \quad v_t^2 = -\frac{2}{\tau} \mathbb{E}_t^{\mathbb{Q}}[\log X_t] \quad (\text{A.5})$$

Under common scale, the premium is exactly proportional to the scale parameter. For small dispersion and suitable moment and integrability conditions, the log-contract expansion gives $v_t \simeq s_t \sqrt{\mathbb{E}^{\mathbb{Q}}[Z^2]/\tau}$ when $X_t = 1 + s_t Z$. Neither relationship identifies the common shape. Different distributions can share the same premium and probability, and their shapes can change while preserving these moments and an approximately proportional movement in v_t .

In a disaster mixture, changing disaster frequency and the dispersion of other outcomes can affect all three objects. The broad shortfall probability need not rise with disaster frequency, because normalization to the forward also changes the location of other outcomes relative to the threshold.

The local Gaussian benchmark. Suppose $X = 1 + \mu + \sigma\epsilon$ under $\mathbb{P}$, where $\sigma > 0$ and $\epsilon \sim N(0, 1)$. With $\theta = \mu/\sigma$, let $w = \exp(-\theta\epsilon - \theta^2/2)$. The normal moment-generating function gives $\mathbb{E}^{\mathbb{P}}[w] = 1$. Moreover,

$$\phi(\epsilon) \exp(-\theta\epsilon - \theta^2/2) = \phi(\epsilon + \theta)$$

Thus ϵ has distribution $N(-\theta, 1)$ under $\mathbb{Q}$, and X has distribution $N(1, \sigma^2)$. It follows that

$$q = \frac{1}{2} \quad a = \frac{\sigma}{\sqrt{2\pi}} \quad m = m^+ = \sigma \sqrt{\frac{2}{\pi}}$$

Under $\mathbb{P}$, the shortfall event is $\{\epsilon < -\theta\}$. The truncated-normal expectation $\mathbb{E}[\epsilon | \epsilon < -\theta] = -\phi(\theta)/\Phi(-\theta)$ gives

$$p = \Phi(-\theta) \quad m^{\mathbb{P}} = \sigma \left[\frac{\phi(\theta)}{\Phi(-\theta)} - \theta \right] \quad \kappa = \frac{q}{p} = \frac{1}{2\Phi(-\theta)}$$

This example distinguishes physical from priced shortfall components under an explicit change of measure.

The equality of the priced and physical variances depends on the exponential-affine change of measure in this benchmark. Gaussian primitive innovations alone do not establish Gaussian equity payoffs or an exponential-affine projected pricing kernel. Nonlinear policies and valuation can change the shape under either measure. The arithmetic-Gaussian construction is also only a local approximation for equity, since it permits negative terminal prices.

A.2.5 VOLATILITY INNOVATIONS: ESTIMATION, RECOVERY, AND SAMPLE SENSITIVITY

Innovation construction and monthly option measures. The first stage uses the original restricted four-lag VAR of Berger et al. (2020). The variable ordering is log realized volatility, log implied volatility, the federal funds rate, log manufacturing employment, and log industrial production. The two volatility equations exclude lags of the three macroeconomic variables. The estimation sample is February 1983–December 2014. I retain the authors' innovation construction and scale realized volatility to match the cumulative future-volatility effect of news over months 1–23, as in their replication code. The component estimates extend these innovations with local projections; they do not use the authors' VAR response estimator.

The option horizon is 30 days. For the final three trading sessions $\mathcal{T}_M$ of month M , let $D_d = e^{-d_d \tau}$ be the daily carry factor and χ_d the DPP. The monthly objects are

$$A_M = \frac{\sum_{d \in \mathcal{T}_M} \chi_d}{\sum_{d \in \mathcal{T}_M} D_d} \quad q_M = \frac{\sum_{d \in \mathcal{T}_M} D_d q_d}{\sum_{d \in \mathcal{T}_M} D_d} \quad m_M = \frac{A_M}{q_M}$$

All three sessions must have usable observations. An earlier session never replaces a missing required observation. The baseline retains 296 of the 300 months in 1990–2014. September 1992, May 1993, August 1994, and October 1994 are excluded. Each daily shortfall is defined relative to that date's forward, so the monthly responses incorporate changes in the forward threshold.

Local projections and origin counts. At each horizon $h = 0, \dots, 24$, the dependent variable is $\log z_{t+h} - \log z_{t-1}$. Regressors are both innovations, an intercept, and four lags of log realized volatility, log implied volatility, the funds rate, employment growth, and industrial-production growth. Component observations must be valid at $t-1$, t , and $t+h$. Macroeconomic projections use the same origin dates and regressors, with cumulative log employment or industrial production as the outcome. Identical samples and regressors preserve the response identity $R_A^j(h) = R_q^j(h) + R_m^j(h)$ for each innovation j , including within every bootstrap draw.

The baseline has 291 impact origins, 275 at twelve months, and 264 at 24 months. Relative to impact, the twelve-month projection loses the twelve origins in January–December 2014 because their endpoints exceed the December 2014 sample boundary. It also loses September 1991, May 1992, August 1993, and October 1993 because the corresponding $t+12$ option components are missing. These four endpoints are September 1992, May 1993, August 1994, and October 1994, respectively. Reporting horizons were fixed after the component experiment and before the macroeconomic estimates were computed.

Bootstrap inference. Inference uses 999 circular-block resamples with 24-month blocks. Resampling supplies frequency weights to intact observations at their original dates, so leads and lags never cross artificial block joins. Each draw re-estimates the first-stage VAR, reconstructs the innovations, and refits all projections using the same weights. The tables include uncertainty in the realized-volatility normalization. The saved fixed-normalization responses are multiplied by the ratio of that draw's normalization to the original normalization. This changes response intervals but leaves component response ratios unchanged.

Pointwise intervals are bootstrap percentiles. The simultaneous bands use the bootstrap maximum absolute deviation from the original estimates, standardized by each estimate's bootstrap standard deviation, over the stated family. They are not separately studentized bootstrap- t intervals. Inference conditions on the recovered daily option components. Comparisons across recovery methods assess sensitivity without incorporating option-recovery uncertainty into these bands.

Upside component responses. Define $m_M^+ = A_M / (1 - q_M)$ on the same monthly observations. The local projections add $\log(1 - q_M)$ and $\log m_M^+$ as outcomes, retaining the original innovations, regressors, masks, and joint bootstrap. The identity

$$R_A^j(h) = R_{1-q}^j(h) + R_{m^+}^j(h)$$

holds at every horizon and in every retained draw. Adding these outcomes reproduces the original component and macroeconomic point estimates and all 999 original joint draws to numerical precision.

Table A.2: Upside impact responses to volatility innovations

Innovation	Gain probability $1 - q$	Gain severity m^+
Realized volatility	0.15 [-0.45, 1.01]	8.61 [3.81, 16.73]
Future-volatility news	0.66 [-0.20, 1.45]	10.78 [9.13, 11.88]

Notes: Entries are 100 times cumulative log responses on impact, using 291 origins. Brackets are 95 percent pointwise percentile intervals from the same 999 joint 24-month circular-block draws. The first stage, innovations, and volatility normalization are re-estimated in every draw. Inference conditions on the recovered option series.

Both innovations increase conditional priced gains as well as losses. The finding concerns two-sided conditional magnitudes; it does not identify an exclusively downside shock or establish invariant shape.

Comparison at equal impact severity. Let $R_z^j(h)$ denote the estimated cumulative log response of outcome z to innovation $j \in \{\text{RV}, \text{news}\}$ under the equal-future-volatility normalization. Scaling innovation j to produce a 0.01 increase in impact log severity gives the macroeconomic response

$$\tilde{R}_z^j(h) = 0.01 \frac{R_z^j(h)}{R_m^j(0)} \quad z \in \{E, Y^{\text{IP}}\} \quad (\text{A.6})$$

Here E is manufacturing employment and Y^{IP} is industrial production. Each bootstrap draw uses its own impact severity denominator, which is positive for both innovations in all 999 draws. With the volatility normalization re-estimated within each draw, the minimum denominator is 0.01057 for realized volatility and 0.09057 for news, in log units. Re-estimating the initial volatility normalization does not change these ratios, because numerator and denominator receive the same multiplier.

The reported contrast is $100[\tilde{R}_z^{\text{RV}}(h) - \tilde{R}_z^{\text{news}}(h)]$. Its simultaneous band covers both macroeconomic contrasts over horizons 0–24, using the same maximum-standardized-deviation construction as the other macroeconomic bands.

A common proportional mapping from impact severity to employment would require

$$R_E^j(h) = \gamma_h R_m^j(0) \quad \text{for both } j \in \{\text{RV}, \text{news}\} \quad (\text{A.7})$$

It would therefore imply equal normalized employment responses. At twelve months, their difference is -0.065

in units of 100 times log changes, with a pointwise interval of $[-0.098, -0.020]$ and a simultaneous interval of $[-0.124, -0.006]$. This ratio-based result is qualified by the division-free check below. Both formulations provide pointwise evidence against the proportional restriction; their simultaneous conclusions differ at twelve months. The comparison does not hold later severity responses fixed, estimate a causal mediation effect, or test a general sufficiency claim for both at-the-money components or their paths.

Division-free inference for the proportional restriction. Write $e_j(h) = R_E^j(h)$ and $b_j = R_m^j(0)$ for cumulative log responses, with $j \in \{R, N\}$. For nonzero impact severities, the common-mapping null is

$$e_R(h)/b_R = e_N(h)/b_N \iff e_R(h)b_N - e_N(h)b_R = 0.$$

For comparable reporting units, the point estimate and bootstrap statistic are

$$\hat{T}_h = \frac{\hat{e}_R(h)\hat{b}_N - \hat{e}_N(h)\hat{b}_R}{\hat{b}_R\hat{b}_N}, \quad T_h^* = \frac{e_R^*(h)b_N^* - e_N^*(h)b_R^*}{\hat{b}_R\hat{b}_N}. \quad (\text{A.8})$$

Because $100 \times 0.01 = 1$, these statistics are in the same units as the reported employment contrast for a 0.01 impact log-severity increase. The denominator is fixed at its point-estimate value in every draw; it sets reporting units and introduces no random division.

Table A.3: Twelve-month employment restriction: two inference formulations

Formulation	Estimate	Pointwise 95% interval	Simultaneous 95% interval
Ratio	-0.065	$[-0.098, -0.020]$	$[-0.124, -0.006]$
Cross-product, fixed scale	-0.065	$[-0.104, -0.023]$	$[-0.1305, 0.0005]$

Notes: Both procedures use the same 999 joint draws and 275 origins at twelve months. Simultaneous bands cover employment and industrial-production restrictions over horizons 0–24, using the maximum standardized bootstrap deviation. Pointwise intervals are percentile intervals. Inference conditions on recovered components.

The point estimates of b_R and b_N are 0.09033 and 0.12379 in log units. Every retained draw has positive impact severities, so the cross-product and ratio have identical signs within each draw. Their bootstrap magnitudes and therefore simultaneous critical values differ. The cross-product simultaneous interval narrowly includes zero at twelve months. The result has pointwise support under both formulations, while simultaneous significance depends on the formulation. The additional check does not hold subsequent severity paths or other equilibrium responses fixed.

Let s_i be the bootstrap standard deviation at coordinate i and define $M_b = \max_{i \in \mathcal{F}} |(T_{b,i}^* - \hat{T}_i)/s_i|$ for the 50-coordinate family $\mathcal{F}$ (employment and industrial production, each at horizons 0–24). The 95 percent critical value is the empirical 0.95 quantile of M_b . For coordinate i , the reported adjusted p -value is

$$p_i^{\text{adj}} = \frac{1 + \sum_{b=1}^{999} \mathbf{1}\{M_b \geq |\hat{T}_i|/s_i\}}{1000}.$$

The ratio critical value is 2.80964 and its twelve-month employment statistic is 3.08809, giving $p^{\text{adj}} = 0.024$. The cross-product values are 3.14906, 3.12350, and 0.053. These are coordinate-wise family-adjusted p -values, not global tests of the entire path. The original reporting protocol listed horizons 0, 6, 12, and 24 and a mean over horizons 0–6; it did not specify a sole primary twelve-month employment hypothesis. No smaller outcome family is selected after inspecting these results.

Two-component mapping and equal-premium comparison. With two innovations, an unrestricted mapping $R_E^j(12) = \beta_q R_q^j(0) + \beta_m R_m^j(0)$ exactly fits the two employment responses whenever the component-response matrix is nonsingular. The point estimates are $\beta_q = -1.414$ and $\beta_m = -0.0953$, with ratio 14.83. Across the joint draws, their central 95 percent ranges are $[-9.77, 12.76]$, $[-0.503, 0.619]$, and $[-60.10, 82.05]$, respectively. The determinant changes sign in 276 of 999 draws; inversion becomes unstable as the compositions approach one another. No draw is trimmed. These descriptive ranges do not identify causal component effects, and an exactly fitting two-weight mapping supplies no additional restriction with which to test component sufficiency.

Replacing $R_m^j(0)$ in (A.6) with $R_A^j(0)$ gives the response to an innovation scaled to raise impact log premium by 0.01. The twelve-month employment contrast is -0.06764 in units of 100 times log changes, with a 95% pointwise interval of $[-0.09960, -0.02166]$ and a simultaneous interval of $[-0.12718, -0.00810]$. This exploratory band covers both macroeconomic contrasts over horizons 0–24 under the premium normalization. It is a separate family from the severity-normalized bands. Impact premium denominators are positive in all 999 draws. The same evidence therefore disfavors a common proportional mapping from the impact premium response alone to twelve-month employment.

Direct comparison of severity paths. Table A.4 reports selected horizons of $100[R_m^{\text{news}}(h) - R_m^{\text{RV}}(h)]$ under the original equal-future-volatility normalization, re-estimated in every draw. The difference changes sign and no horizon from 0 through 24 excludes zero even pointwise. The simultaneous band covers this one contrast over all 25 horizons, separately from the macroeconomic bands. This exploratory comparison was added after inspecting the original results; it establishes neither a larger or more persistent severity response to news nor equivalence of the paths. The reporting window remains 0–24 and the initial volatility normalization uses months 1–23.

Table A.4: Severity responses and their difference

Horizon	RV	News	News minus RV		
			Difference	Pointwise	Simultaneous
0	9.03	12.38	3.35	$[-6.41, 7.96]$	$[-8.80, 15.49]$
6	3.67	2.86	-0.82	$[-5.14, 4.24]$	$[-8.29, 6.66]$
12	2.62	1.85	-0.78	$[-6.39, 3.95]$	$[-9.48, 7.93]$
24	1.17	2.10	0.93	$[-4.22, 6.28]$	$[-7.88, 9.73]$

Notes: Entries are 100 times cumulative log-severity responses. Both innovations retain the equal-future-volatility normalization, with the RV multiplier re-estimated in each of the 999 joint draws. Intervals refer to the news-minus-RV difference and are 95%. The simultaneous band covers all horizons 0–24, including those not displayed. Origins number 291, 281, 275, and 264 at the four displayed horizons. Differences are formed before rounding. This is an exploratory comparison; an insignificant difference does not establish equivalence. Recovery uncertainty is not included in these bands.

Recovery methods and sample sensitivity. The recovery exercise uses the put-spread baseline, call spreads, a smooth curve, and entropy recovery with lognormal and uniform reference distributions. Each method retains its own fitted premium. I estimate responses on each baseline–alternative intersection and on the common intersection of all five methods. Across these comparisons, the impact severity intervals remain strictly positive for both innovations.

Table A.5 shows how the associated employment contrast changes with the available dates under the equal-future-volatility normalization. The call and smooth samples preserve coverage, and the simultaneous contrast remains negative. The entropy and five-method intersections lose substantial coverage, especially in 1992–1997, and yield wider bands. Because the macroeconomic projections do not use recovered components as regressors, this table measures sample sensitivity of the macroeconomic contrast.

Table A.5: Twelve-month employment contrast across recovery samples

Sample	Months	Origins	Contrast	Pointwise	Simultaneous
Original baseline	296	275	−0.628	[−0.929, −0.224]	[−1.159, −0.097]
Baseline–call	292	263	−0.645	[−0.953, −0.209]	[−1.201, −0.090]
Baseline–smooth	295	273	−0.627	[−0.922, −0.208]	[−1.159, −0.094]
Entropy: lognormal	180	123	−0.687	[−1.226, 0.010]	[−1.992, 0.618]
Entropy: uniform	181	125	−0.713	[−1.387, 0.025]	[−2.050, 0.624]
All five methods	178	118	−0.681	[−1.255, 0.083]	[−2.059, 0.698]

Notes: RV-minus-news manufacturing-employment responses, multiplied by 100, with normalization re-estimated in every draw. Origins refer to $h = 12$. Each entropy row uses its intersection with the baseline. Intervals are 95%. Each simultaneous band covers both macroeconomic contrast paths over horizons 0–24 within that sample; coverage is not joint across the six samples. Pairings were fixed before estimation and all are reported. Baseline, call, and smooth samples retain 999 draws; entropy pairings retain 994 and the five-method sample retains 993. Rank-deficient resamples are recorded in the replication output and excluded jointly within each comparison.

The equal-impact-severity employment contrast also remains negative on the high-coverage intersections. Using baseline severity on the call-spread intersection gives -0.067 in units of 100 times log changes at twelve months, with a 95% simultaneous interval of $[-0.127, -0.006]$. Using call-spread severity gives -0.069 and $[-0.133, -0.005]$. On the smooth-curve intersection, baseline severity gives -0.064 and $[-0.123, -0.006]$; smooth-curve severity gives -0.063 and $[-0.120, -0.007]$. Each band covers both macroeconomic contrasts over horizons 0–24 within its sample and severity normalization. Coverage is not joint across samples, methods, or normalizations. All four comparisons use 999 retained joint draws, with positive impact severity denominators in every draw. These added comparisons are exploratory. Equal-severity comparisons on the lower-coverage entropy intersections are not reported here.

An additional comparison examines the RV-minus-news difference in impact probability-to-premium ratios on the identical five-method sample. The differences are 0.004 for put spreads, 0.071 for call spreads, 0.084 for the smooth curve, 0.127 for entropy with a lognormal reference, and 0.111 for entropy with a uniform reference. Only the lognormal-reference contrast excludes zero under a simultaneous band covering all five method-specific differences. Its interval is $[0.016, 0.237]$, compared with $[-0.002, 0.223]$ for the uniform reference. This comparison was added after inspecting earlier results. Adjustment across methods does not make it a prespecified test, and the evidence does not establish a recovery-robust distinction between the two innovation classes' composition ratios.⁷

A.3 SUPPLEMENT TO THE DPP DECOMPOSITION This appendix supplies the derivations, component-recovery details, stability comparisons, and quotation calculations supporting Section 3. It separates sampling inference conditional on recovered components from the range of decompositions allowed by quotation intervals.

A.3.1 DECOMPOSITION AND CONDITIONAL LOSS–GAIN ASYMMETRY Let $X = S_{t+\tau}/F_{t,t+\tau}$, so that $\mathbb{E}_t^Q[X] = 1$. The adjusted premium is $a = \mathbb{E}_t^Q[(1 - X)^+] = qm$. The premium alone does not identify its components, even after imposing the forward-mean restriction and positive terminal values. For any $a \in (0, 1)$ and $q \in (a, 1)$, assign probability q to $X = 1 - a/q$ and probability $1 - q$ to $X = 1 + a/(1 - q)$. Both values are positive, the mean is one, and the expected shortfall is a , while the shortfall probability can take any such q . For a strictly positive terminal value, $0 < m < 1$ whenever a shortfall has positive probability, so $a < q$.

⁷Differences in fitted premium responses contribute modestly to the entropy ratios. Dividing the entropy news-probability responses by the baseline premium response gives -0.114 and -0.116 , compared with their own-premium ratios of -0.117 and -0.120 . This arithmetic comparison does not replace each method's internally consistent decomposition.

Probability levels and severity asymmetry. Since $\mathbb{E}_t^Q[X_t - 1] = 0$,

$$\mathbb{E}_t^Q[(1 - X_t)^+] = \mathbb{E}_t^Q[(X_t - 1)^+]$$

Define $m_t^+ = \mathbb{E}_t^Q[X_t - 1 \mid X_t \geq 1]$. For $0 < q_t < 1$, conditioning on the two events gives

$$q_t m_t = (1 - q_t) m_t^+ \quad \frac{m_t}{m_t^+} = \frac{1 - q_t}{q_t}$$

These give the two identities in (5). The identity remains valid with an atom at the forward when that atom is included in the conditioning event defining m_t^+ . For a continuous distribution, $q_t = 1/2$ places the forward at a median. It does not imply symmetry. Conversely, symmetry about the forward implies $q_t = 1/2$ when there is no atom there.

The ratios evaluated at $q = 0.423$ and $q = 0.444$ are approximately 1.364 and 1.252. These calculations transform the reported group mean probabilities. They are not group means of the date-specific severity ratios. For any common set of averaging weights,

$$\mathbb{E} \left[\frac{m_t}{m_t^+} \right] = \mathbb{E} \left[\frac{1}{q_t} \right] - 1 \geq \frac{1}{\mathbb{E}[q_t]} - 1 \quad (\text{A.9})$$

provided the expectation exists. The inequality follows from convexity of $1/q$. Differences in the dispersion of q_t across groups can affect the mean ratio, so an ordering of group mean probabilities does not by itself establish the reverse ordering of group mean severity ratios.

A.3.2 SCALE BENCHMARKS AND LOCAL OPTION INFORMATION Proposition 1 supplies an exact additive common-scale benchmark. The following derivations distinguish it from lognormal scale variation and a fixed nonlinear payoff map.

The lognormal benchmark. Let $X = \exp(-s^2/2 + s\varepsilon^Q)$, with $s > 0$ and $\varepsilon^Q \sim N(0, 1)$. Then $\mathbb{E}^Q[X] = 1$ and $X < 1$ exactly when $\varepsilon^Q < s/2$. Therefore

$$q = \Phi(s/2) \quad \mathbb{E}^Q[X \mathbf{1}_{\{X < 1\}}] = \Phi(-s/2)$$

Subtracting the second expression from the first gives $a = 2\Phi(s/2) - 1$. Consequently,

$$q = \frac{1+a}{2} \quad m = \frac{2a}{1+a} \quad m^+ = \frac{2a}{1-a} \quad \frac{m}{m^+} = \frac{1-a}{1+a} \quad (\text{A.10})$$

The associated within-family elasticities satisfy

$$\frac{d \log q}{d \log a} = \frac{a}{1+a} \quad \frac{d \log m}{d \log a} = \frac{1}{1+a} \quad (\text{A.11})$$

These are analytical elasticities within the lognormal family. They are not estimates of the aggregate covariance loading in Section 3.

For small a , severity accounts for nearly all premium variation. Nevertheless, $q > 1/2$ and $m/m^+ < 1$ for every $s > 0$. A probability below one half cannot be obtained by changing the volatility parameter within this family. The lognormal family is not an exact additive common-scale family around the forward: both q and standardized shape change with s .

The local implied-volatility slope. At a fixed maturity, write normalized strike as $k = K/F$ and total implied standard deviation as $\omega(\ell)$, where $\ell = \log k$. The normalized Black–Scholes representation of a put price is

$$\mathcal{P}(k) = k\Phi(-d_2) - \Phi(-d_1), \quad d_1 = \frac{-\log k}{\omega} + \frac{\omega}{2}, \quad d_2 = d_1 - \omega.$$

This represents implied volatility, without assuming that the actual priced distribution is lognormal. At $k = 1$, the price is $a = 2\Phi(\omega(0)/2) - 1$. For a differentiable arbitrage-free price curve, with no atom at the forward, total differentiation gives

$$q = \mathcal{P}'(1) = \Phi(\omega(0)/2) + \phi(\omega(0)/2)\omega'(0).$$

Define the premium-adjusted local probability gap by

$$\Delta_t^\tau \equiv q_t^\tau - \frac{1 + a_t^\tau}{2}. \quad (\text{A.12})$$

The preceding derivative yields

$$\Delta_t^\tau = \phi\left(\frac{\omega_t(0)}{2}\right) \frac{\partial \omega_t(\ell)}{\partial \ell} \Big|_{\ell=0}. \quad (\text{A.13})$$

This is a single-maturity identity. The finite-strike estimator approximates the local derivative; maturity and temporal mixtures preserve the premium decomposition but need not have the implied-volatility slope of an exactly traded target-horizon claim. The gap is distinct from global third-moment skewness.

Fixed payoff curvature.

Proposition 3 (Fixed curvature and the probability response). *Let ε have a symmetric, bounded-support distribution under $\mathbb{Q}$, with $\mathbb{E}[\varepsilon^2] = 1$ and a twice continuously differentiable density f near zero, where $f(0) > 0$. Write $\mu = \mathbb{E}[(-\varepsilon)^+]$. For a fixed coefficient c and sufficiently small $\sigma > 0$, define*

$$X_\sigma = 1 + \sigma\varepsilon + \frac{c}{2}\sigma^2(\varepsilon^2 - 1). \quad (\text{A.14})$$

Then X_σ is positive and has priced mean one. Its probability, premium, and local probability elasticity satisfy

$$q(\sigma) = \frac{1}{2} + \frac{cf(0)}{2}\sigma + O(\sigma^3), \quad a(\sigma) = \mu\sigma + O(\sigma^3), \quad (\text{A.15})$$

$$\eta_q(\sigma) \equiv \frac{d \log q(\sigma)}{d \log a(\sigma)} = \frac{q(\sigma) - 1/2}{q(\sigma)} + O(\sigma^3). \quad (\text{A.16})$$

For $c < 0$, sufficiently small increases in scale both deepen conditional losses and lower the shortfall probability.

Proof. Bounded support ensures positivity and monotonicity in ε for sufficiently small σ . Symmetry and unit variance give $\mathbb{E}[X_\sigma] = 1$. The threshold e_σ satisfying $X_\sigma = 1$ solves $e_\sigma + (c\sigma/2)(e_\sigma^2 - 1) = 0$, hence $e_\sigma = c\sigma/2 + O(\sigma^3)$. Symmetry gives $f'(0) = 0$, so $q(\sigma) = \Pr(\varepsilon < e_\sigma) = 1/2 + cf(0)\sigma/2 + O(\sigma^3)$. In the shortfall integral, the coefficient of σ^2 over $\varepsilon < 0$ vanishes because $\mathbb{E}[(\varepsilon^2 - 1)\mathbf{1}\{\varepsilon < 0\}] = 0$. Moving the integration boundary from zero to e_σ adds $O(\sigma^3)$, yielding $a(\sigma) = \mu\sigma + O(\sigma^3)$. Differentiation gives $q'(\sigma) = cf(0)/2 + O(\sigma^2)$ and $a'(\sigma) = \mu + O(\sigma^2)$, which imply (A.16). For $c < 0$, $q' < 0$ and $a' > 0$ locally, so $m = a/q$ rises. $\square$

The proposition is local and concerns a fixed distribution under the pricing measure. It does not assert that the conditional law generated by a general second-order equilibrium solution belongs to this family. Nor does a local

elasticity equal the regression coefficient over a finite sample unless additional conditions make the relationship log-linear over the relevant range.

For the exact monthly mappings in Table A.6, I retain all 439 observed A_M values, define each counterfactual q_M observation by observation, and re-estimate the same loading on 438 adjacent changes and the same recession gap on 36 recession and 403 expansion months. The constant-gap calibration uses $\delta = 439^{-1} \sum_M [q_M - (1 + A_M)/2]$. No distribution for quotation errors is imposed. These counterfactual statistics are diagnostics conditional on observed premiums; the table does not provide a joint sampling test for calibrated model parameters.

Finite-change benchmark diagnostics. The probability level and recession comparison distinguish restrictions that a near-unit severity loading alone cannot. A single common shape spanning the full sample implies constant q and therefore $\mathcal{G}(q) = 0$, conflicting with the positive estimated gap. An exactly lognormal priced distribution implies $q > 1/2$ on every date, conflicting with the recovered phase means below one-half. Holding the premium-adjusted local gap $\Delta = q - (1 + A)/2$ constant also imposes a testable restriction. Its estimated monthly recession increase is 1.391 percentage points, with a 95 percent HAC interval of $[0.258, 2.524]$, inconsistent with a constant gap. These comparisons condition on component recovery.

For a finite-change comparison, I apply each exact probability mapping to the observed monthly adjusted premiums and use the same regressions and recession dates. The resulting probability loadings and log gaps are reported in Table A.6. This calculation avoids replacing a sample regression coefficient by a local derivative evaluated at a sample mean.

Table A.6: Probability restrictions and finite-change benchmark diagnostics

Mapping evaluated on monthly premiums	β_q	$\mathcal{G}(q)$
Observed components	-0.032	0.049
Common scale: constant q	0.000	0.000
Lognormal: $q = (1 + A)/2$	0.022	0.014
Constant gap: $q = (1 + A)/2 + \delta$	0.026	0.016

Notes: Mappings are evaluated on the original 439 monthly adjusted premiums, with 438 adjacent changes and 36 recession months. The constant gap δ equals the sample mean of $q - (1 + A)/2$. The table reports descriptive model-implied statistics, not confidence intervals or tests treating calibrated parameters as known. The exact level and phase restrictions discussed in the text supply the corresponding benchmark comparisons.

Fixed curvature gives a different local restriction. At $q = 0.425$, its leading-order elasticity is approximately -0.176 ; at $q = 0.409$ it is approximately -0.222 . These are local illustrations, not the implied monthly or quarterly regression coefficients. Even with one varying parameter, a nonlinear probability mapping generally makes the covariance loading depend on the entire path of premiums. A comparison with fixed curvature therefore requires finite-scale predictions and control of approximation error. The present calculation does not reject perturbation methods or all one-parameter priced-distribution families. The global fixed-shape comparison also differs from the within-month quotation test in Section 3.5, which permits a different candidate shape in every month.

A.3.3 COMPONENT RECOVERY AND AGGREGATION

Identification and convexity bounds. For a fixed maturity, write $P(K) = e^{-r\tau} \mathbb{E}^Q[(K - S_{t+\tau})^+]$. The put payoff is Lipschitz in K , so differentiating under the expectation at a strike without an atom gives $P'(K) = e^{-r\tau} \mathbb{Q}(S_{t+\tau} < K)$. Evaluating at the forward proves Proposition 2. If there is an atom, the left derivative prices the strict shortfall event, while the right derivative also includes the atom.

Convexity gives, for $K_- < F < K_+$,

$$e^{r\tau} \frac{P(F) - P(K_-)}{F - K_-} \leq q \leq e^{r\tau} \frac{P(K_+) - P(F)}{K_+ - F}. \quad (\text{A.17})$$

The two expressions are averages of the cumulative probability over the left and right intervals, respectively. For positive probability bounds $\underline{q}$ and $\bar{q}$, the corresponding severity bounds are $a/\bar{q}$ and $a/\underline{q}$. If the AtM price is linearly interpolated from these same two quoted prices, the two secants coincide by construction. That coincidence does not establish that the underlying derivative is known. The quotation calculations instead permit quoted prices to vary within their bid–ask intervals, conditional on the constructed AtM premium and original forward.

Maturity and monthly aggregation. Let w be the original weight on the longer component maturity, with carry factors D_s, D_ℓ and adjusted premiums a_s, a_ℓ . Define

$$D = (1 - w)D_s + wD_\ell, \quad q = \frac{(1 - w)D_s q_s + wD_\ell q_\ell}{D}, \quad m = \frac{\chi}{Dq}. \quad (\text{A.18})$$

Then $\chi = Dqm$ exactly, and $a = \chi/D$ is the carry-adjusted mixture of the two component premiums. These weights preserve the synthetic daily premium and its decomposition; they do not identify a separately traded target-maturity law.

Let $\mathcal{D}_M$ contain the common daily observations in month M , with n_M dates. Since $\chi_j = D_j q_j m_j$, define

$$\bar{D}_M = \frac{1}{n_M} \sum_{j \in \mathcal{D}_M} D_j, \quad q_M = \frac{\sum_{j \in \mathcal{D}_M} D_j q_j}{\sum_{j \in \mathcal{D}_M} D_j}, \quad m_M = \frac{\sum_{j \in \mathcal{D}_M} D_j q_j m_j}{\sum_{j \in \mathcal{D}_M} D_j q_j}. \quad (\text{A.19})$$

Thus $\bar{\chi}_M = \bar{D}_M q_M m_M$ and $A_M = q_M m_M$. With constant carry, q_M is an arithmetic daily average, while m_M remains a probability-weighted severity average. The same construction preserves $A_M = (1 - q_M)m_M^+$ on identical dates. The monthly components describe a mixture of daily forward-looking distributions, not the distribution of a realized calendar-month return.

Sample and admissibility. The original maturity and quotation selection is retained before checking whether estimates are admissible. A selected maturity must have $0 < \hat{q} < 1$, $a > 0$, and $a \leq \hat{q}$, with a numerical tolerance of 10^{-12} on the last inequality. Missing brackets or failures of these conditions exclude 34 of the 8,825 premium dates (0.39 percent), including one recession date. No truncation is applied and the retained daily premiums are unchanged. All 439 months remain, with 8,791 daily component observations and 438 adjacent monthly changes. Restricting monthly averages to component dates gives correlations with the original DPP of 0.99998 in levels and 0.99984 in log changes; the carry-adjusted correlations are 0.99998 and 0.99982.

Empirical attribution and inference. On identical observations, the log-product identity implies $\beta_q + \beta_m = 1$ for regressions with intercepts, and $\mathcal{G}(A) = \mathcal{G}(q) + \mathcal{G}(m)$ for phase log gaps. The two component regressions have opposite residuals and equal slope standard errors. These covariance contributions need not lie between zero and one. Baseline intervals use Bartlett Newey–West covariance with six monthly lags and the finite-sample adjustment $n/(n - 2)$. Recession shares use the delta method with the joint covariance of the component and premium log gaps. The severity-minus-probability recession gap is 0.477, with a 95 percent interval of [0.276, 0.679]. All intervals condition on recovered components; quotation uncertainty is assessed separately below.

A.3.4 TIME STABILITY, TWO-SIDED EVIDENCE, AND OPTION HORIZONS This subsection reports the time-stability and recession comparisons for the baseline put-spread decomposition. The sample contains 439 months from January 1990 through July 2026 and 438 adjacent monthly changes. The adjusted premium and its components satisfy $A_M = q_M m_M$ in every month, with identical daily observations used to construct all three monthly series. Severity is the risk-neutral expected proportional shortfall below the forward, conditional on that event.

Rolling elasticities Table A.7 reports rolling estimates of the severity elasticity from regressions of $\Delta \log m_M$ on an intercept and $\Delta \log A_M$. The corresponding probability regression uses the same observations, so the two slope estimates sum to one in each window. Severity elasticities range from 0.951 to 1.159 in five-year windows, from 0.987 to 1.097 in ten-year windows, and from 1.005 to 1.064 in fifteen-year windows. Figures A.6 and A.7 show the five- and fifteen-year estimates; the ten-year estimates appear in Figure 4 in the main text and are summarized in Table A.7.

Confidence intervals use a Bartlett Newey–West covariance estimator with six monthly lags and the finite-sample adjustment $n/(n-2)$, where n is the number of observations in the regression. They are conditional on the recovered component series. The windows overlap, and the intervals are pointwise rather than simultaneous bands. The lowest lower confidence limit is therefore a descriptive summary of these intervals, not a simultaneous confidence bound or a formal test that the coefficients are constant through time.

Table A.7: Severity Elasticities Across Rolling Windows

Window	Windows	Min. $\hat{\beta}_m$	Max. $\hat{\beta}_m$	Lowest pointwise 95% lower limit
5 years	379	0.951	1.159	0.869
10 years	319	0.987	1.097	0.935
15 years	259	1.005	1.064	0.957

Notes: Windows contain 60, 120, or 180 adjacent monthly changes. Their first endpoints are January 1995, January 2000, and January 2005, respectively; all end by July 2026. Each regression includes an intercept. The last column reports the minimum of the pointwise 95 percent lower confidence limits across windows, using six monthly HAC lags. These limits do not provide simultaneous coverage.

Sensitivity to the HAC lag length Table A.8 compares three, six, and twelve monthly HAC lags in the full sample. Changing the lag length affects the covariance estimates, leaving the point estimates unchanged. The severity elasticity remains close to one, the probability elasticity remains close to zero, and the confidence interval for the recession contribution contrast $\mathcal{G}(m) - \mathcal{G}(q)$ remains strictly positive. Here $\mathcal{G}(z)$ is the mean log level of z_M in recession months minus its mean log level in expansion months, and the severity share is $100\mathcal{G}(m)/\mathcal{G}(A)$.

Table A.8: Sensitivity of Conditional Inference to the HAC Lag Length

Statistic	Estimate	95% confidence interval		
		3 lags	6 lags	12 lags
Severity elasticity $\hat{\beta}_m$	1.032	[0.995, 1.070]	[0.994, 1.071]	[0.993, 1.071]
Probability elasticity $\hat{\beta}_q$	−0.032	[−0.070, 0.005]	[−0.071, 0.006]	[−0.071, 0.007]
Probability gap $\mathcal{G}(q)$	0.049	[0.023, 0.075]	[0.021, 0.077]	[0.021, 0.077]
Contribution contrast $\mathcal{G}(m) - \mathcal{G}(q)$	0.477	[0.292, 0.662]	[0.276, 0.679]	[0.276, 0.679]
Severity share (%)	91.5	[86.7, 96.3]	[86.7, 96.3]	[86.8, 96.1]

Notes: Elasticities use 438 adjacent monthly changes; recession gaps and shares use 439 monthly levels, including 36 recession and 403 expansion months. All regressions include an intercept and use Bartlett Newey–West covariance estimates with the $n/(n-2)$ adjustment. Contrast intervals retain the covariance between component estimates. Share intervals use the delta method and the joint covariance of the severity and premium gaps. All intervals use normal critical values and are conditional on the recovered component series.

Recession dating and omitted episodes Table A.9 compares the baseline recession definition with peak-inclusive dating and with the omission of each recession in turn. The baseline assigns months after each business-cycle peak through the trough to recession; peak months belong to expansions. Peak-inclusive dating transfers the four peak months to recessions, raising the severity share from 91.5 to 91.7 percent. The omission comparisons remove only the recession months of the specified episode and retain all 403 baseline expansion months. Severity shares range from 90.0 to 93.3 percent, and every reported 95 percent confidence interval for $\mathcal{G}(m) - \mathcal{G}(q)$ lies above zero. The pooled comparison is therefore not driven by any one of the four recessions in this sample.

Table A.9: Recession Definitions and Leave-One-Recession-Out Comparisons

Comparison	N_R/N_E	$\mathcal{G}(A)$	$\mathcal{G}(q)$	$\mathcal{G}(m)$	Severity share (%)	95% CI for $\mathcal{G}(m) - \mathcal{G}(q)$
Baseline	36/403	0.576	0.049	0.527	91.5	[0.276, 0.679]
Peak inclusive	40/399	0.545	0.045	0.500	91.7	[0.268, 0.641]
Omit 1990–91	28/403	0.640	0.055	0.585	91.3	[0.295, 0.763]
Omit 2001	28/403	0.620	0.041	0.578	93.3	[0.310, 0.764]
Omit 2007–09	18/403	0.461	0.040	0.420	91.3	[0.155, 0.605]
Omit 2020	34/403	0.547	0.055	0.492	90.0	[0.236, 0.638]

Notes: N_R and N_E are the numbers of recession and expansion months in each comparison. $\mathcal{G}(z)$ is the recession-minus-expansion mean log gap on the indicated sample. Shares are computed from unrounded gaps as $100\mathcal{G}(m)/\mathcal{G}(A)$. The omission rows retain baseline recession dating and remove the indicated recession months from the sample, without reclassifying them as expansion months. Confidence intervals use joint six-lag HAC covariance estimates, the $n/(n-2)$ adjustment, and normal critical values. Calendar-month lag distances are preserved after omission; observations on opposite sides of an omitted episode are not treated as adjacent months.

Local expansion benchmarks Table A.10 replaces the full expansion sample with the twelve or twenty-four months immediately preceding each recession, including the peak month. For each component, the local gap is its mean log level during the recession minus its mean log level in the available expansion reference months. The table reports these gaps as well as the severity share, so that the magnitude of the premium gap remains visible.

Severity accounts for 72.5–98.9 percent of these local premium gaps. The 2001 comparison is sensitive to the reference period: the severity share is 98.8 percent using twelve preceding months and 72.5 percent using twenty-four, with premium log gaps of only 0.058 and 0.082, respectively. Only seven preceding months are available for the 1990–91 recession, so both requested reference windows use those same seven months. These episode comparisons are descriptive and do not include episode-specific confidence intervals.

Monthly Upside Components and Local Asymmetry For each of the original 439 monthly observations, define $m_M^+ = A_M/(1 - q_M)$ and $\Delta_M = q_M - (1 + A_M)/2$. All transformations precede averaging and estimation. The upside identity $A_M = (1 - q_M)m_M^+$ therefore holds observation by observation. The local gap is the mixture counterpart of the single-maturity object; it need not be the slope of an exactly traded 30-day implied-volatility curve.

Gain severity averages 3.07 percent in expansions and 5.72 percent in recessions. The observation-level loss-to-gain ratios average 1.373 and 1.258, respectively. Conditional gains rise more proportionately, while conditional losses remain larger in phase averages. The identity $\mathcal{G}(m) - \mathcal{G}(m^+) = \mathcal{G}((1 - q)/q)$ separates relative conditional magnitudes from the premium's overall movement.

The 30-day monthly gap averages -0.08569 in expansions and -0.07178 in recessions. Its increase is 1.391 percentage points, with a six-lag HAC interval of $[0.258, 2.524]$ percentage points. The premium term adds 0.704 percentage points to the 2.095-percentage-point probability increase. This local asymmetry calculation is distinct from global third-moment skewness.

Table A.10: Recession Log Premiums Relative to the Preceding Expansion

Recession	N_R/N_E	$\mathcal{G}_{\text{local}}(A)$	$\mathcal{G}_{\text{local}}(q)$	$\mathcal{G}_{\text{local}}(m)$	Severity share (%)
<i>Panel A: Preceding twelve months</i>					
1990–91*	8/7	0.229	0.008	0.221	96.6
2001	8/12	0.058	0.001	0.058	98.8
2007–09	18/12	0.702	0.097	0.606	86.3
2020	2/12	1.228	0.036	1.192	97.1
<i>Panel B: Preceding twenty-four months</i>					
1990–91*	8/7	0.229	0.008	0.221	96.6
2001	8/24	0.082	0.023	0.060	72.5
2007–09	18/24	0.846	0.070	0.776	91.7
2020	2/24	1.211	0.013	1.198	98.9

Notes: N_R is the number of recession months and N_E is the number of available expansion reference months. Recession intervals are August 1990–March 1991, April–November 2001, January 2008–June 2009, and March–April 2020. Peak months belong to the preceding expansion.

*The January 1990 sample start leaves only seven preceding months for the first recession, so neither requested reference window is complete. Shares equal $100\mathcal{G}_{\text{local}}(m)/\mathcal{G}_{\text{local}}(A)$, calculated from unrounded mean log differences. No episode-specific confidence intervals are reported.

Table A.11: Monthly upside and relative-severity evidence

Object	Estimate	95% interval
Gain-probability loading	0.023	[−0.003, 0.050]
Gain-severity loading	0.977	[0.950, 1.003]
Gain-probability recession log gap	−0.037	[−0.058, −0.015]
Gain-severity recession log gap	0.612	[0.392, 0.833]
Loss-to-gain ratio recession log gap	−0.086	[−0.135, −0.037]

Notes: Loadings use 438 adjacent monthly changes. Recession gaps use 439 months, including 36 recession months. Regressions, observations, six-lag Bartlett HAC inference, and the correction $n/(n-2)$ match the baseline downside analysis. Intervals condition on recovered components.

Table A.12: Quarterly-horizon probability and conditional severity asymmetry

Method	Probability q		Loss-to-gain severity m/m^+	
	Expansion	Recession	Expansion	Recession
Put spread	0.408 [0.396, 0.420]	0.422 [0.400, 0.435]	1.471 [1.401, 1.545]	1.385 [1.304, 1.530]
Call spread	0.410 [0.398, 0.421]	0.430 [0.399, 0.443]	1.458 [1.390, 1.528]	1.338 [1.259, 1.518]
Smooth curve	0.408 [0.397, 0.420]	0.428 [0.398, 0.444]	1.461 [1.391, 1.534]	1.353 [1.252, 1.537]

Notes: Option horizon: 91.25 ACT/365 days. Entries are phase means on 136 common quarter-end observations (124 expansion and 12 recession). Each observation combines the final three exchange sessions, all valid under all methods. The smooth curve has a free AtM anchor. Severity-ratio entries average observation-level $(1 - q)/q$; they are not ratios evaluated at average probabilities. Brackets give 95 percent point-wise percentile intervals from 1,999 circular calendar-block draws with eight-quarter blocks, conditional on recovery. Missing quarters remain in the sampling frame. Quarter-end phase uses the final month's NBER status, with peaks in expansions and troughs in recessions. Sample: 1990Q1–2026Q2.

Quarterly-horizon evidence The quarterly construction uses 91.25 ACT/365 days, exactly one quarter of a year. It retains the 30-day construction's quotation filters, parity forward, and strike and maturity interpolation rules. Put-spread, call-spread, and smooth-curve estimates use the same selected chains, with maturities bracketing the target.

Each observation combines the final three exchange sessions of a completed calendar quarter. All three must have valid components; invalid sessions are left missing. Common dates are selected across methods before carry-weighted aggregation, preserving $a = qm = (1 - q)m^+$. The resulting sample has 136 quarter-end observations: 124 in expansions and 12 in recessions, spanning four recession episodes. Completed-quarter comparisons end in 2026Q2.

The mean put-spread premium is 3.55 percent of the forward, with a 95 percent interval of [3.15, 3.97] percent. Expansion and recession means are 3.36 and 5.51 percent. These levels provide a horizon-matched target separately from the monthly 30-day series.

Expansion probabilities range from 0.408 to 0.410 across recoveries (Table A.12). Their observation-level conditional loss-to-gain ratios average 1.458–1.471: conditional priced losses are 46–47 percent larger than gains. Recession probabilities range from 0.422 to 0.430, with mean ratios of 1.338–1.385. The probability and ratio columns report the same underlying local information in probability and conditional-loss units.

The level is more precisely established than the phase change. The put-spread recession-minus-expansion probability difference is 0.014, with an interval of [−0.009, 0.030]; the ratio difference is −0.086, with an interval of [−0.186, 0.060]. All three recoveries have phase-difference intervals containing zero. Alternative sampling conventions give similar probability levels: means of 0.412–0.413 on 9,080 common daily dates and 0.410–0.413 on each method's own quarter-end sample.

The lognormal gap separates the probability level from the premium contribution. The mean quarterly put-spread gap is −0.109, with an interval of [−0.120, −0.098]; call-spread and smooth-curve means are −0.106 and −0.108. A lognormal volatility chosen to match each observed premium leaves this probability discrepancy by construction.

The quarterly local-gap change is 0.318 percentage points, with an eight-quarter calendar-block interval of [−1.819, 1.758] percentage points. Table A.13 compares it with the monthly 30-day result, then aligns both horizons on identical quarter-end dates. The aligned construction uses the same final three exchange sessions for each horizon and retains 134 quarters, including 12 recession quarters. Its recession gap changes are 2.393 percentage points at 30 days and 0.250 at 91 days. Their directly estimated difference is 2.144 percentage points, with a joint

Table A.13: Local asymmetry across phases and option horizons

Construction	Expansion $100\bar{\Delta}_E$	Recession $100\bar{\Delta}_R$	Change R minus E	95% interval for change
A. Original observation rules				
30 days, monthly averages	-8.569	-7.178	1.391	[0.258, 2.524]
91 days, quarter-end mixture	-10.892	-10.574	0.318	[-1.819, 1.758]
B. Identical quarter-end dates at both horizons				
30 days	-8.969	-6.576	2.393	[-0.495, 4.832]
91 days	-10.824	-10.574	0.250	[-1.881, 1.695]
30 minus 91 days	1.855	3.998	2.144	[0.598, 4.057]

Notes: All entries are percentage points of $\Delta = q - (1 + a)/2$. Panel A uses 439 monthly observations at 30 days and 136 quarter-end observations at 91.25 ACT/365 days. Its monthly interval uses six-lag HAC; quarterly intervals use eight-quarter circular calendar blocks. Panel B uses baseline put spreads on 134 common quarters, with both horizons combining exactly the same final three exchange sessions; 12 quarters are in recessions. Differences are formed jointly in 1,999 attempted calendar-block draws. Five draws contain no recession observations and are omitted jointly, leaving 1,994 valid draws. Missing calendar quarters remain in the resampling frame. Intervals are pointwise and conditional on recovery.

Table A.14: Quarterly probability loadings on protection-premium changes

Recovery	Quarters	Changes	β_q	95% HAC interval
Exact final exchange session				
Put spread	142	137	0.0064	[-0.0798, 0.0926]
Call spread	142	137	0.0182	[-0.0707, 0.1072]
Smooth curve	142	137	0.0486	[-0.0005, 0.0976]
Final-three-session mixture				
Put spread	136	128	0.0358	[-0.0292, 0.1008]
Call spread	136	128	-0.0226	[-0.1139, 0.0688]
Smooth curve	136	128	0.0438	[-0.0027, 0.0903]

Notes: All options have a 91.25 ACT/365-day target. Recovery methods use common valid observations within each sampling convention; exact-date and three-session panels have their own eligible quarters. OLS includes an intercept. Bartlett HAC uses four actual calendar-quarter lags and the finite-sample adjustment $n/(n-2)$. Missing quarters are not bridged. Intervals condition on recovered components. Severity loadings equal $1 - \beta_q$. On the identical 136-quarter comparison sample, the exact-date smooth estimate is 0.0517 with interval [0.0034, 0.1000]; this sensitivity is retained in the accompanying results.

interval of [0.598, 4.057]. The covariance retained by the paired comparison makes this difference more precise than either phase change alone.

The local asymmetry comparison is therefore horizon dependent: the recession change is larger at the shorter horizon on the aligned sample. This evidence concerns a vega-weighted local implied-volatility slope. It does not equate the local gap to a standardized third moment or interpret an imprecise quarterly change as exact acyclical-ity.

The same construction permits a matched-frequency comovement comparison. Table A.14 reports the probability loading using either the exact final exchange session or the final-three-session mixture. Only changes between adjacent calendar quarters enter. The exact-date baseline loading is 0.0064, with a complementary severity loading of 0.9936. The three-session baseline gives 0.0358 and 0.9642. The six probability intervals include zero, and the point estimates imply severity loadings between 0.9514 and 1.0226. These results extend severity dominance to quarterly changes; they do not establish equality of the monthly and quarterly coefficients or an exactly constant probability.

Exact-date observations are the direct empirical counterpart of a quarterly model's date-specific conditional distribution. Three-session observations retain the paper's averaging convention. The exact-date baseline has

mean premium 3.5167 percent, mean probability 0.41123, and mean local gap -0.10636 , compared with 3.5485 percent, 0.40911, and -0.10864 in the mixture. A model comparison should choose its corresponding observation rule before assessing a discrepancy.

Component levels and alternative rolling windows Figure A.5 reports the monthly adjusted premium, shortfall probability, and conditional severity in their natural units. Its vertical scales differ across panels; proportional variation is compared on a common scale in the main text. The rolling figures use equal vertical ranges centered on the severity and probability benchmarks of one and zero, respectively. They provide a visual account of the rolling estimates summarized in Table A.7.

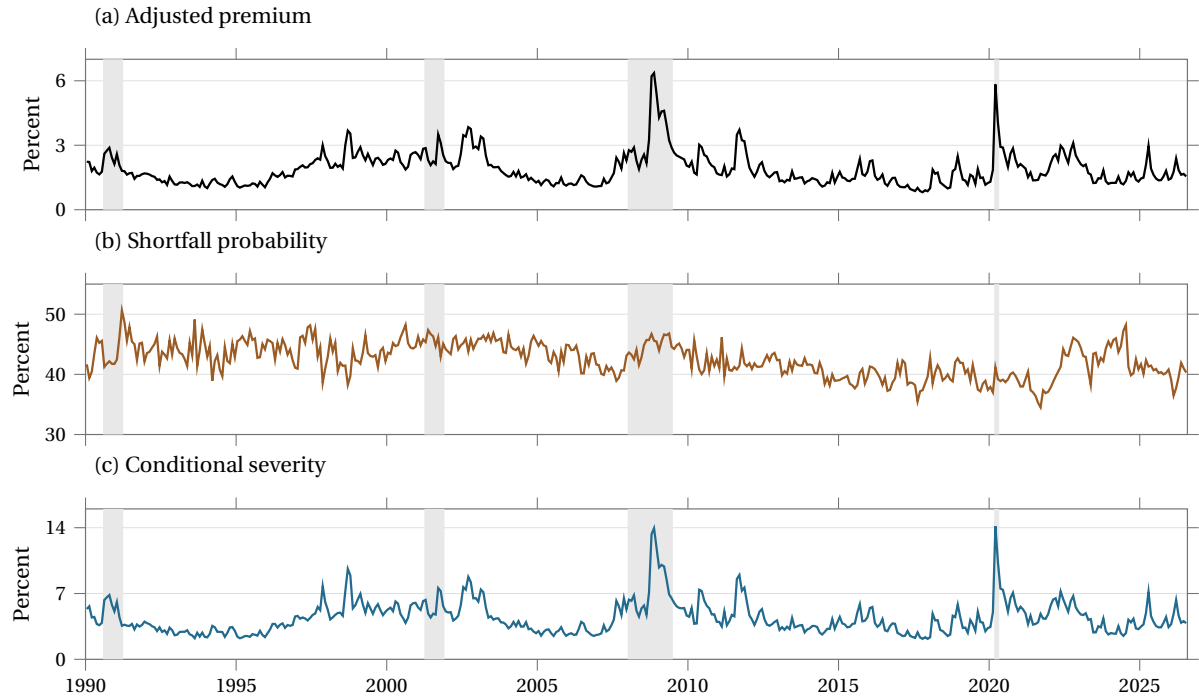


Figure A.5: Monthly Premium, Probability, and Severity in Natural Units

Notes: The panels show $100A_M$, $100q_M$, and $100m_M$, respectively, from January 1990 through July 2026. Severity is expressed as a percentage of the forward value. Vertical scales differ across panels. Gray shading denotes recession months, from the month after the business-cycle peak through the trough. Lines are baseline recovered estimates without component-level uncertainty bands.

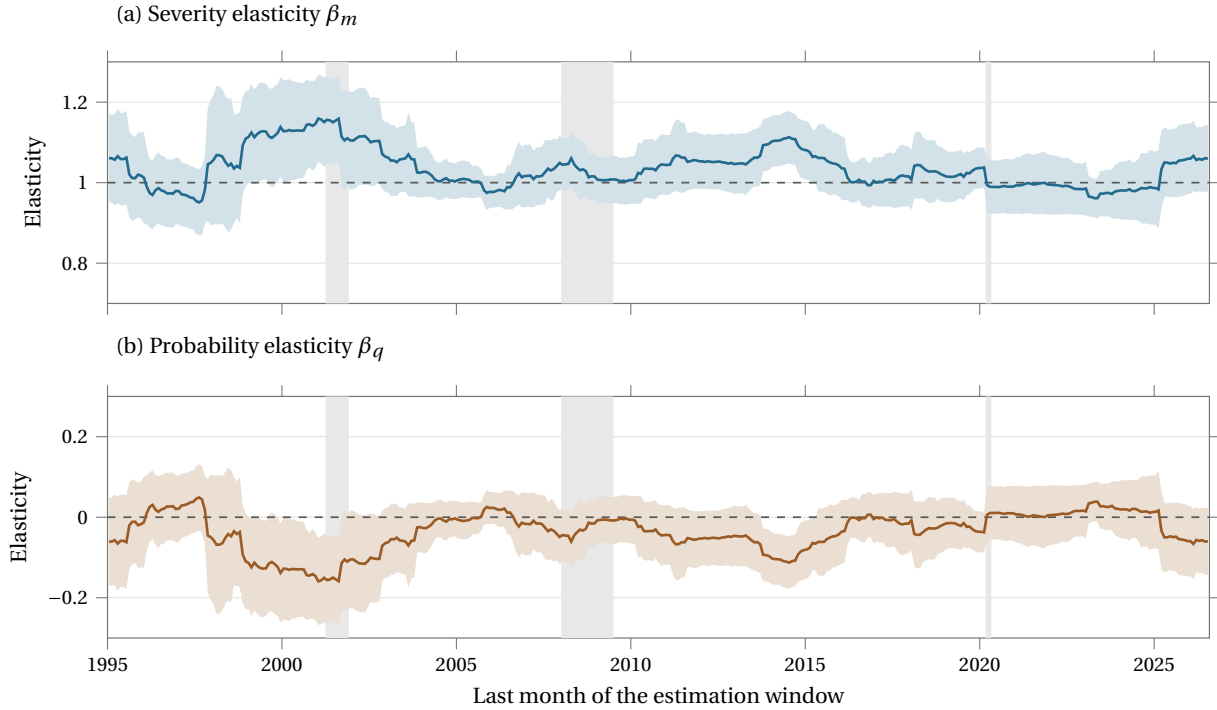


Figure A.6: Probability and Severity Elasticities in Rolling Five-Year Windows

Notes: Each window contains 60 adjacent monthly changes, with endpoints from January 1995 through July 2026. Lines are slope estimates from regressions with an intercept; shaded bands are pointwise 95 percent intervals using six-lag Bartlett Newey–West covariance estimates and the $n/(n-2)$ adjustment, conditional on the recovered series. The two panels use equal vertical ranges centered on one and zero, with dashed lines marking these benchmarks. Gray recession shading refers to the window endpoint, not to the composition of the entire estimation window. Overlapping windows do not provide simultaneous coverage.

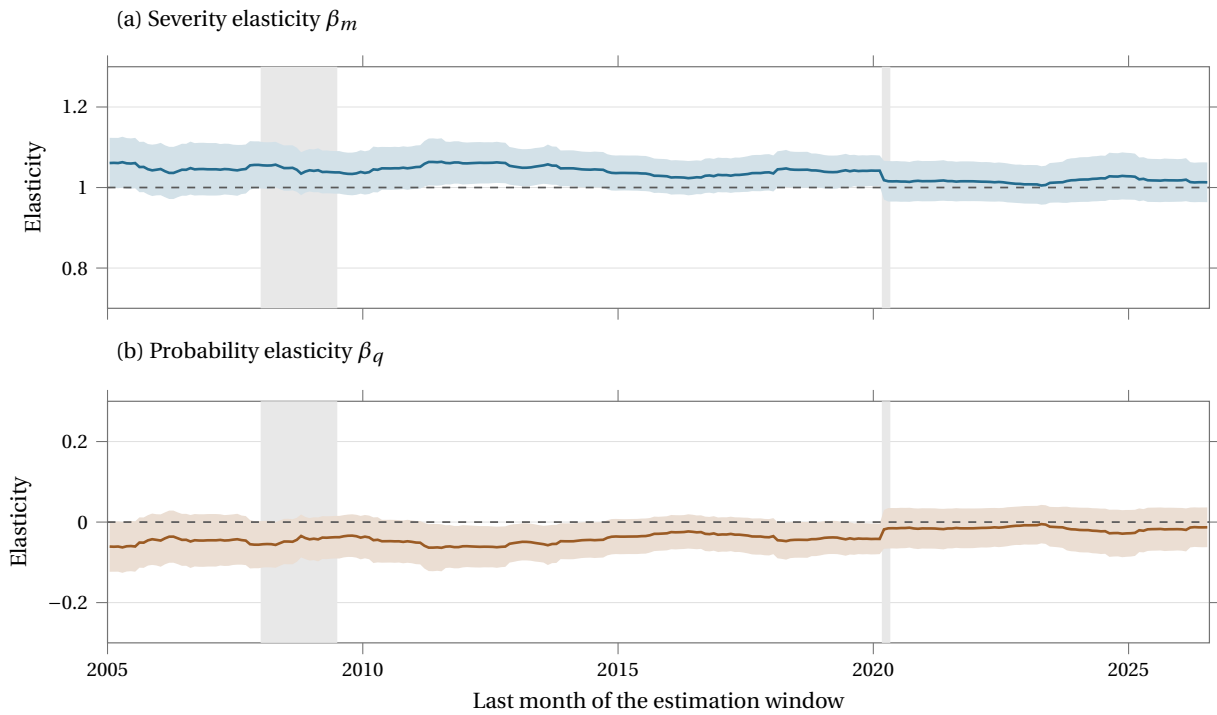


Figure A.7: Probability and Severity Elasticities in Rolling Fifteen-Year Windows

Notes: Each window contains 180 adjacent monthly changes, with endpoints from January 2005 through July 2026. Regression construction, pointwise 95 percent intervals, benchmark lines, equal vertical ranges, and endpoint recession shading follow Figure A.6.

A.3.5 ROBUSTNESS OF THE PROBABILITY–SEVERITY DECOMPOSITION The analysis distinguishes uncertainty about the recovered components from sampling uncertainty conditional on those components. Alternative recovery methods and quotation bounds assess sensitivity to the former; HAC inference and the joint block bootstrap address the latter. Every recovery comparison retains the original forward, snapshot, settlement classification, and selected maturities. Each method is compared with the baseline on identical usable daily dates before monthly aggregation, with coverage and failures reported. Agreement across recoveries does not provide independent protection against quotation errors shared by the methods. Each recovery retains assumptions about interpolation, support, or regularization, and all comparisons condition on the original forward and maturity selection.

The full baseline contains 8,791 daily component observations and 439 months from January 1990 through July 2026. For each comparison, I first intersect usable daily dates and then rebuild all monthly series using the carry-adjusted aggregation in Appendix A.3.3. The resulting monthly series satisfy $A_M^{(j)} = q_M^{(j)} m_M^{(j)}$ for method j . Monthly log changes use adjacent calendar months only; missing months are not bridged. The severity elasticity is the OLS coefficient from regressing $\Delta \log m_M^{(j)}$ on $\Delta \log A_M^{(j)}$ with an intercept. The recession severity share is $\mathcal{G}(m^{(j)})/\mathcal{G}(A^{(j)})$, where each gap is the recession mean log level minus the expansion mean log level. Recessions run from the month after an NBER peak through the trough.

Unless otherwise indicated, confidence intervals use Newey–West covariance with a Bartlett kernel, six calendar-month lags, and the finite-sample correction $n/(n-2)$. Missing calendar months contribute zero regression scores to the HAC calculation; lag distances are not compressed. Share intervals use the delta method and the joint covariance matrix of the component and premium log gaps. All such intervals are conditional on the recovered component series.

Alternative recovery methods I compare the nearest put spread with wider strike brackets, quote-quality restrictions, and call-side spreads. The quality checks require positive bids, a relative bid–ask spread at most 25 percent, or volume of at least five; a separate check requires a strike bracket no wider than one percent of the forward. These restrictions operate within the original eligible quotation set. Call-side spreads provide a quotation-consistency check because call and put slopes are linked by put–call parity. For a call bracket $K^- < F < K^+$, the corresponding estimate is

$$\hat{q}^{\text{call}} = 1 + e^{r_t u} \frac{C(K^+, u) - C(K^-, u)}{K^+ - K^-}. \quad (\text{A.20})$$

The nearest call bracket and calls at the baseline put strikes are examined separately. Their agreement is a consistency check on quotations linked by put–call parity; it does not provide an independent source of pricing information.

For the curve and entropy fits, write normalized strike as $k = K/F$ and normalized put price as $p(k) = e^{r_t u} P(Fk, u)/F$. The selected quotations are valid, non-crossed puts within $|k - 1| \leq 0.20$, retaining the nearest strict strike on each side of one and requiring at least five strikes. Duplicate strikes must be resolved before recovery.

The smooth-curve estimator represents the risk-neutral distribution using nonnegative masses on return intervals, with a uniform density within each interval. The implied put-price curve is convex and continuously differentiable. Mass and mean are constrained to one, and the curve is fitted to midpoint prices using quotation-spread weights and a curvature penalty. The default fit uses at most 21 nearby put strikes within 20 percent of the forward; 11 and 41 strikes provide window sensitivities. The curvature penalty is also varied by a factor of ten in either direction. Because the fit does not require every fitted price to lie inside its quotation interval, pricing residuals and band violations are reported.

More precisely, let w_j be the mass on interval $[b_{j-1}, b_j]$, let $d_j = b_j - b_{j-1}$, and define its average put payoff by

$$H_j(k) = \frac{1}{d_j} \int_{b_{j-1}}^{b_j} (k - x)_+ dx, \quad p_w(k) = \sum_j w_j H_j(k). \quad (\text{A.21})$$

The restrictions are $w_j \geq 0$, $\sum_j w_j = 1$, and $\sum_j w_j (b_{j-1} + b_j)/2 = 1$. Given N quotations with midpoint prices p_i^{mid} , the smooth fit minimizes

$$\frac{1}{2N} \sum_{i=1}^N \left[\frac{\tilde{h}}{h_i} \frac{p_w(k_i) - p_i^{\text{mid}}}{s} \right]^2 + \frac{\lambda}{2} \sum_j \frac{w_j^2}{d_j}, \quad (\text{A.22})$$

where h_i is the normalized quotation half-spread with a 10^{-6} numerical floor, $\tilde{h}$ is its cross-quote median, and $s = \max\{a^{\text{baseline}}, 10^{-5}\}$. The default penalty is $\lambda = 10^{-5}$; the sensitivities use 10^{-6} and 10^{-4} . The penalty is the integrated squared density, equivalently the integrated squared curvature of the implied put-price curve.

The entropy estimator instead minimizes relative entropy over the interval masses subject to probability normalization, unit mean, and fitted prices inside every selected bid–ask interval. Lognormal and uniform reference distributions provide prior sensitivity; the fitted probabilities remain free to change at every date. The main support is $[10^{-6}, 3]$, and a wider, refined specification uses $[10^{-8}, 10]$. Bounds obtained by optimizing over these intervals are conditional on the specified density family and support. With positive reference masses π_j , the entropy objective is $\sum_j w_j \log(w_j/\pi_j)$, subject to the mass and mean constraints and $p_i^{\text{bid}} \leq p_w(k_i) \leq p_i^{\text{ask}}$ for every selected quote. The reference is normalized over the chosen support, with a 10^{-14} floor on reference masses before normalization; fitted masses and shortfall probabilities are not floored. The interval endpoints contain the selected strikes, one, fixed tail knots, and 33 lognormal-spaced interior knots. The lognormal scale $\nu = 2\Phi^{-1}((1 + a^{\text{baseline}})/2)$ locates the interior knots and defines the lognormal reference. It does not restrict the fitted distribution to be lognormal. The wider specification uses 65 interior knots, includes the main grid and its interval midpoints, and extends the support as stated above.

For both approaches, I distinguish fits that hold the original adjusted premium fixed from fits that estimate it jointly with probability. This distinction matters because the original premium is constructed by linear interpolation: imposing that premium and both surrounding midpoints on a convex curve can mechanically reproduce the original local slope. The joint fits produce their own coherent premium, probability, and severity, and the difference between the fitted and original premium is reported. The fitted interval masses and shortfall probability are allowed to change at every date, within the stated support and density family. All fits are date-specific; no temporal smoothing is imposed. Successful recovery requires normalized equality residuals at most 2×10^{-8} and mass negativity no greater than 10^{-10} ; entropy fits additionally satisfy quotation bands to within 2×10^{-8} . The smooth solver checks a global convex optimality-gap bound against 10^{-7} ; entropy fits require a primal–dual gap in $[-10^{-8}, 10^{-6}]$. Infeasible problems and failed numerical certificates are excluded and recorded rather than assigned component estimates.

Results on common dates Table A.15 compares the baseline put spread, the call spread, the smooth curve, and the two entropy reference distributions on exactly the same 7,078 daily dates. This sample contains 408 usable months, 398 adjacent monthly changes, and all 36 recession months. The smooth and entropy estimates in this table jointly fit the premium and probability, so agreement does not require holding the original premium fixed.

The severity elasticities range from 1.016 to 1.055, and severity accounts for 90.8–93.5 percent of the recession log premium. The smooth estimate is 1.042, with a 95 percent HAC interval of $[1.009, 1.076]$; its recession severity share is 91.9 percent, with an interval of $[87.0, 96.9]$ percent. Probability’s monthly log-change standard deviation ranges from 0.032 to 0.056 across the five methods, compared with 0.184–0.197 for severity. Thus, the result

Table A.15: Component Recovery on Identical Daily Samples

Method	Severity elasticity	Recession severity share (%)	$sd(\Delta \log q_M)$	$sd(\Delta \log m_M)$
<i>Panel A: All common usable dates</i>				
Baseline put spread	1.039 [0.997, 1.081]	91.2 [86.1, 96.3]	0.056	0.195
Call spread	1.055 [1.017, 1.092]	92.0 [87.2, 96.9]	0.054	0.197
Smooth curve, joint premium	1.042 [1.009, 1.076]	91.9 [87.0, 96.9]	0.034	0.191
Entropy, lognormal reference	1.016 [0.972, 1.060]	90.8 [86.0, 95.6]	0.032	0.184
Entropy, uniform reference	1.027 [0.985, 1.069]	93.5 [89.1, 97.9]	0.033	0.186
<i>Panel B: At least ten common dates per month</i>				
Baseline put spread	1.038 [0.997, 1.080]	90.5 [85.2, 95.7]	0.041	0.196
Call spread	1.036 [1.006, 1.065]	91.4 [86.1, 96.6]	0.039	0.195
Smooth curve, joint premium	1.038 [1.004, 1.071]	91.9 [86.9, 96.8]	0.031	0.194
Entropy, lognormal reference	1.011 [0.966, 1.056]	90.4 [85.4, 95.4]	0.030	0.187
Entropy, uniform reference	1.022 [0.979, 1.065]	93.4 [88.9, 97.8]	0.030	0.189

Notes: Panel A uses 7,078 daily dates, 408 months, 398 adjacent monthly changes, and 36 recession months. Panel B uses 6,895 daily dates, 361 months, 355 adjacent changes, and 32 recession months. The dates are identical across all five methods within each panel. Brackets contain 95 percent HAC confidence intervals, using a Bartlett kernel, six calendar-month lags, and the correction $n/(n-2)$. Recession shares are $100\mathcal{G}(m)/\mathcal{G}(A)$; their intervals use the delta method and the joint covariance of the log gaps. Each smooth or entropy method uses its own jointly fitted premium. Standard deviations refer to the same adjacent changes used to estimate elasticities. Inference is conditional on the recovered component series.

concerns both comovement with the premium and comparatively limited probability variation. Several probability elasticities are slightly negative, so the findings support near-unit severity elasticity without establishing that probability is constant or that distributional shape is fixed.

Panel B of the table requires at least ten common dates in every retained month, leaving 6,895 dates, 361 months, and 355 adjacent changes, including 32 recession months. Severity elasticities remain between 1.011 and 1.038, and recession severity shares remain between 90.4 and 93.4 percent. The wider-bracket, quotation-quality, smoothing-window, curvature-penalty, and entropy-support comparisons also retain severity dominance. Each of these specifications is evaluated against a baseline rebuilt on its own common daily sample; those samples need not coincide with the five-method sample in Table A.15.

Tables A.16 and A.17 report the additional specifications and the baseline recalculated on each specification's matched sample. The positive-bid and one-percent-bracket checks can operate primarily through sample selection, so their agreement with the matched baseline should not be read as an independent change in the recovery method.

Coverage and pricing diagnostics Table A.18 reports coverage and fitted-premium differences. These diagnostics qualify the method comparison. The jointly fitted smooth estimator recovers components on 8,812 of the 8,825 premium dates, including 8,779 with baseline components. On the latter sample its premium differs from the baseline by -0.28 percent on average, and the 95th percentile of the absolute relative difference is 1.75 percent. Thirty-seven dates have absolute relative differences above five percent, with a maximum of 40.1 percent. As an ex-post influence diagnostic, excluding these dates and rebuilding both series on the remaining dates leaves the smooth severity elasticity at 1.040 and its recession severity share at 92.7 percent, compared with 1.039 and 92.7 percent before exclusion. This diagnostic does not replace the main estimates or establish that all fitted prices lie within their quotation bands.

The median smooth-fit root mean squared pricing residual is 0.116 quotation half-spreads, and its 95th percentile is 1.439 half-spreads, calculated across valid maturity-level fits. The largest recorded smooth optimality-gap bound is 8.3×10^{-9} , below the acceptance threshold, and the largest equality residual is 2.9×10^{-15} . These numerical certificates concern optimization and the moment restrictions; the smooth objective does not impose hard quotation bands.

The jointly fitted entropy methods recover 7,099 and 7,101 dates under the lognormal and uniform references, respectively. Coverage is particularly sparse in the early sample: the lognormal reference retains only five dates in 1993, one in 1994, and four in 1995. Most failures reflect infeasible quotation bands within the specified density family. Common-sample comparisons address differences in observation dates; they do not recover information on dates where an entropy fit is unavailable. On their respective baseline-matched samples, the fitted premiums differ from the baseline by 0.98 and 1.26 percent on average; the corresponding 95th percentiles of absolute relative differences are 3.07 and 3.76 percent.

Outer bounds from option quotations For each selected maturity, I bound $q = \mathbb{Q}(X < 1)$ over a linear relaxation of positive-return distributions whose mean is one, whose at-the-money put price equals the original constructed premium, and whose put prices lie inside all retained bid-ask quotations. The relaxation retains both probability mass and the first moment above the largest quoted strike, so it imposes no finite upper return support. It also permits limiting endpoint configurations; consequently, the intervals are conservative outer bounds rather than a claim that every endpoint is attained by a strictly positive distribution. An atom at the forward is treated separately from the strict shortfall event. The relaxation retains every eligible put quotation at the selected maturity, rather

Table A.16: Spread and Quotation-Quality Comparisons

Specification	Daily dates	Severity elasticity	Severity share (%)	Matched baseline elasticity; share (%)
Second-nearest put bracket	8,753	1.045 [1.013, 1.078]	93.0 [88.7, 97.3]	1.033; 91.4
Third-nearest put bracket	8,708	1.060 [1.025, 1.094]	93.3 [88.6, 98.1]	1.034; 91.5
Positive put bids	8,788	1.033 [0.994, 1.071]	91.5 [86.7, 96.3]	1.033; 91.5
Relative spread $\leq 25\%$	8,747	1.030 [0.992, 1.069]	91.5 [86.9, 96.2]	1.030; 91.5
Put volume ≥ 5	8,418	1.046 [1.009, 1.082]	92.7 [88.0, 97.4]	1.035; 91.6
Bracket width $\leq 1\%$ of forward	6,347	1.023 [0.983, 1.064]	89.3 [83.8, 94.8]	1.023; 89.3
Nearest call bracket	8,760	1.035 [1.006, 1.064]	92.5 [88.0, 97.0]	1.031; 91.4
Calls on the baseline put bracket	8,730	1.035 [1.006, 1.064]	92.6 [88.1, 97.1]	1.031; 91.5

Notes: Each row uses the intersection of that specification's valid daily dates with the baseline, before monthly aggregation. The final column gives the baseline severity elasticity and recession severity share recalculated on those same dates. Samples can differ across rows and from Table A.15. Brackets contain conditional 95 percent HAC intervals with six calendar-month lags; share intervals use the joint delta method. The spread methods retain the original premium. Joint smooth and entropy fits estimate their own premium; fixed-premium fits impose the original value. No missing calendar month is bridged when forming log changes.

Table A.17: Smooth-Curve and Entropy Specifications

Specification	Daily dates	Severity elasticity	Severity share (%)	Matched baseline elasticity; share (%)
Smooth, joint premium	8,779	1.039 [1.008, 1.070]	92.7 [88.0, 97.3]	1.033; 91.4
Smooth, fixed premium	8,779	1.041 [1.010, 1.072]	92.7 [87.9, 97.5]	1.033; 91.5
Smooth, 11 quotes, joint premium	8,780	1.037 [1.006, 1.069]	92.6 [88.0, 97.2]	1.033; 91.5
Smooth, 41 quotes, joint premium	8,779	1.038 [1.008, 1.069]	92.9 [88.4, 97.5]	1.033; 91.4
Smooth, penalty 10^{-6} , joint	8,780	1.035 [1.004, 1.066]	92.2 [87.5, 96.9]	1.033; 91.4
Smooth, penalty 10^{-4} , joint	8,781	1.046 [1.013, 1.078]	93.9 [89.6, 98.1]	1.033; 91.4
Entropy, lognormal, joint premium	7,094	1.017 [0.976, 1.058]	90.8 [86.0, 95.6]	1.043; 91.1
Entropy, uniform, joint premium	7,096	1.030 [0.989, 1.070]	93.5 [89.1, 97.9]	1.043; 91.2
Entropy, lognormal, fixed premium	7,026	1.024 [0.984, 1.064]	91.3 [86.4, 96.3]	1.046; 91.1
Entropy, uniform, fixed premium	7,033	1.036 [0.994, 1.077]	93.6 [89.1, 98.1]	1.045; 91.1
Entropy, wide support, fixed premium	7,038	1.026 [0.983, 1.068]	91.4 [86.5, 96.2]	1.049; 91.1

Notes: Each row uses the intersection of that specification's valid daily dates with the baseline, before monthly aggregation. The final column gives the baseline severity elasticity and recession severity share recalculated on those same dates. Samples can differ across rows and from Table A.15. Brackets contain conditional 95 percent HAC intervals with six calendar-month lags; share intervals use the joint delta method. The spread methods retain the original premium. Joint smooth and entropy fits estimate their own premium; fixed-premium fits impose the original value. No missing calendar month is bridged when forming log changes.

Table A.18: Coverage and Fitted-Premium Differences

Method	Valid dates	Baseline-matched dates	Mean relative difference (%)	Absolute relative difference (%) 95th percentile
Baseline put spread	8,791	8,791	—	—
Call spread	8,787	8,760	—	—
Smooth curve, joint premium	8,812	8,779	−0.28	1.75
Entropy, lognormal reference	7,099	7,094	0.98	3.07
Entropy, uniform reference	7,101	7,096	1.26	3.76

Notes: Each method is attempted on the 8,825 original premium dates. A daily estimate requires recovery at every selected maturity needed for that date. Premium differences are $100(\chi_t^{(j)} / \chi_t^{\text{baseline}} - 1)$, calculated on each method's baseline-matched daily sample. The last column is the 95th percentile of their absolute values. These are relative price differences, not percentage-point changes in the DPP. Dashes indicate that the method retains the original premium. Counts in this table are method-specific; the five-way intersection contains 7,078 dates.

than only the nearby fitting window. It partitions at quoted strikes and at zero and one, retaining both tail mass and its first moment above the largest knot. A separate left-limit copy of one distinguishes mass just below the forward from an atom exactly at it. Infeasible quotation sets are reported without relaxing their bands.

The secant inequalities in (A.17) apply to the true price curve. If the AtM price is linearly interpolated between the same two neighboring strikes, both secant slopes equal the baseline put-spread slope. That mechanical equality provides no independent robustness check. The quotation-bound calculation instead lets each quoted put price vary within its observed bid–ask interval while keeping the original AtM premium fixed.

The original positive carry weights map maturity-level intervals into daily and monthly probability intervals $[L_M, U_M]$. Given the premium series, both empirical attributions are linear functions of $\log q_M$. For monthly comovement, writing $x_M = \Delta \log A_M$ and $h_M = (x_M - \bar{x}) / \sum_{J \in \mathcal{C}} (x_J - \bar{x})^2$ on the set $\mathcal{C}$ of valid adjacent changes gives

$$\beta_q = \sum_{M \in \mathcal{C}} h_M (\log q_M - \log q_{M-1}) = \sum_M c_M \log q_M.$$

Each transition adds h_M to c_M and subtracts it from c_{M-1} . The minimum is obtained by choosing $\log L_M$ when $c_M \geq 0$ and $\log U_M$ when $c_M < 0$; the maximum reverses these choices. The severity interval follows from $\beta_m = 1 - \beta_q$. For the recession comparison, the same calculation uses weights $R_M / n_R - (1 - R_M) / n_E$ and $\mathcal{G}(m) / \mathcal{G}(A) = 1 - \mathcal{G}(q) / \mathcal{G}(A)$ when $\mathcal{G}(A) > 0$. These are exact extrema over the box of monthly intervals and conservative attribution envelopes for the underlying distributions. They describe quotation uncertainty conditional on the original forward and premium, not sampling confidence intervals.

The quotation envelopes are not sufficiently tight to establish severity dominance on their own. On the 5,978 dates common to the distributional outer-bound estimator and the baseline, the severity-elasticity envelope is $[-0.064, 2.165]$, and the recession severity-share envelope is $[20.2, 163.3]$ percent. The former uses 374 adjacent monthly changes and the latter 387 months. The outer envelopes include values below equal probability and severity contributions; this does not establish that such values are attainable by admissible underlying distributions. The empirical conclusion is therefore stability across the reported component-recovery methods, rather than identification from quotation bounds alone.

Table A.19 also reports the convexity-only relaxation and the two finite-density-family calculations. The latter remain conditional on their specified support and interval density representation.

Table A.19: Quotation-Based Attribution Envelopes

Restrictions	Daily dates	Months / changes	Severity-elasticity envelope	Recession severity-share envelope (%)
Convexity inequalities	8,553	439/438	[−0.085, 2.181]	[26.9, 160.9]
Finite density family, main support	7,041	408/398	[−0.060, 2.184]	[22.0, 163.8]
Finite density family, wider support	7,056	409/399	[−0.060, 2.185]	[22.2, 163.6]
Distributional outer relaxation	5,978	387/374	[−0.064, 2.165]	[20.2, 163.3]

Notes: Each row uses the dates common to the stated bound calculation and the baseline, holding the original forward and premium fixed. Months apply to recession comparisons; adjacent changes apply to elasticities. Samples differ across rows, so widths should not be read as comparisons on identical observations. These are measurement envelopes, not sampling confidence intervals. Finite-family results depend on the stated support and interval-density representation; the distributional outer relaxation has no finite upper return bound. Attribution extrema over monthly interval boxes need not be jointly attainable by admissible price curves.

Recovery Error and Regression Attribution Conditional on the constructed premium, write $\log \hat{q}_M = \log q_M + e_M$ and $x_M = \Delta \log A_M$. Then

$$\hat{\beta}_q - \beta_q = \frac{\text{Cov}(\Delta e_M, x_M)}{\text{Var}(x_M)}, \quad \mathcal{G}(\hat{q}) - \mathcal{G}(q) = \mathbb{E}[e_M | R] - \mathbb{E}[e_M | E]. \quad (\text{A.23})$$

The probability loading shifts with $\text{Cov}(\Delta e_M, \Delta \log A_M)$, while the recession gap shifts with the difference in mean log-recovery error across phases. Orthogonality of these log errors preserves the population estimands; unbiased errors in probability levels alone need not do so after taking logs. If the premium is also measured with error, both regressor error and its covariance with probability error enter the loading.

The interval-box extrema choose each $\log q_M$ according to the sign of the attribution weight. They are worst-case choices, not draws from a distribution of quotation noise. Agreement across recovery methods is informative about the implemented estimators but cannot establish orthogonality when the same quotations enter all methods. The HAC and block-bootstrap intervals condition on recovered components and should not be read as incorporating a separate quotation-error model.

The 34 excluded component dates To assess whether the missing component dates could explain the baseline result, I retain the estimated probabilities on the 8,791 usable dates and allow probability on each of the other 34 dates to range over $[a_t, 1]$. This is a conservative closed admissibility range under positive returns and the forward-mean restriction; the limiting endpoints need not be attained. I then rebuild the monthly premium and probability intervals using all 8,825 premium dates and apply the preceding attribution formulas. The resulting severity elasticity lies between 1.003 and 1.058, and severity's share of the recession log premium lies between 90.5 and 92.6 percent. The severity-minus-probability recession log gap remains between 0.466 and 0.490. Thus, the excluded dates cannot overturn severity dominance when the retained component estimates are held fixed. These ranges isolate exclusion sensitivity; they do not incorporate measurement error on the retained dates.

Joint block-bootstrap sensitivity For the original 439-month baseline sample, I assess sampling uncertainty by jointly resampling monthly log components, their original adjacent log changes, and the recession indicator in circular blocks of 6, 12, and 24 months. Each block length uses 4,999 draws, and the table reports marginal 95 percent percentile intervals. Changes are computed before resampling, so concatenating blocks creates no artificial monthly changes; the probability and severity adding-up restrictions continue to hold in every usable draw.

Across the three block lengths, the smallest lower endpoint for the recession severity share is 84.5 percent, and the smallest lower endpoint for the severity-minus-probability recession log gap is 0.255. The largest upper

Table A.20: Joint Block-Bootstrap Sensitivity

Block months	Severity elasticity	Recession severity share (percent)	Probability log gap	Relative probability volatility
6	[0.998, 1.072]	[85.2, 96.8]	[0.018, 0.077]	[0.209, 0.277]
12	[0.996, 1.074]	[84.8, 97.9]	[0.010, 0.077]	[0.203, 0.284]
24	[0.996, 1.076]	[84.5, 98.5]	[0.009, 0.077]	[0.197, 0.297]

Notes: Entries are marginal 95 percent percentile intervals. Relative probability volatility is $\text{sd}(\Delta \log q_M) / \text{sd}(\Delta \log m_M)$. Identical monthly index draws are used for the monthly comovement and recession comparisons. The first original month's unavailable change stays missing when that month is drawn. Draws without both recession and expansion observations are excluded only from the recession calculations; counts are reported in the replication diagnostics. Negative recession-premium draws are retained and shares are not clipped.

endpoint for the volatility of probability relative to severity is 0.297. These comparisons describe the marginal intervals and are not a simultaneous test across block lengths.

These intervals measure sampling sensitivity conditional on the recovered component series. They do not incorporate uncertainty in the recovery of probabilities from option quotations. Only four recession episodes are available; resampling blocks reuses and can split these episodes, and does not produce additional independent cycles. The bootstrap therefore complements the comparisons across time windows and individual recessions, while its finite-sample coverage and sensitivity to structural breaks remain limitations.

A.3.6 COMMON SCALE AND CONDITIONAL DOWNSIDE SHAPE This subsection reports the recovered-probability comparisons and the implementation of the joint quotation restrictions. All calculations use the original 30-day construction, carrying-cost inputs, component at-the-money premiums, and selected quotation bands. The main sample contains 6,966 daily dates in 405 covered calendar months between January 1990 and July 2026. These are not 405 consecutive calendar months. The archived outputs retain daily coverage, corrected scalar dual bounds, numerical residuals, and fitted distributions.

Scale predictions away from the forward. Let $G_t(k) = \mathbb{Q}_t(X_t < k)$. Under common scale, relative to date b ,

$$\lambda_{t,b} = \frac{a_t}{a_b} = \frac{s_t}{s_b}. \quad (\text{A.24})$$

$$G_{t,b}^{\text{scale}}(k) = G_b\left(1 + \frac{k-1}{\lambda_{t,b}}\right). \quad (\text{A.25})$$

The composition is $q_t = \mathbb{Q}(Z < 0)$, $a_t = s_t \mathbb{E}[(-Z)^+]$, and $m_t = s_t \mathbb{E}[-Z \mid Z < 0]$.

$$G_{t,b}^{\text{scale}}(0.95) = G_b(0.975) \quad \text{when } a_t = 2a_b. \quad (\text{A.26})$$

Thus downside probabilities away from the forward can rise even with constant probability at the forward. The scale transformation must preserve positivity: $\mathbb{Q}_b[1 + \lambda_{t,b}(X_b - 1) > 0] = 1$. Replacing negative transformed values by zero changes the mean and premium and does not preserve this benchmark.

Recovered-probability comparisons. For a fixed reference date b , define the discrepancy between the observed probability and the common-scale prediction as

$$\varepsilon_{t,b}(k) = G_t(k) - G_{t,b}^{\text{scale}}(k) \quad (\text{A.27})$$

This is also the difference between the observed probability change from b and the change predicted by the scale benchmark. Predictions are formed at the daily frequency before aggregation, using identical dates and weights for observed and predicted values. An average of daily distributions with different scales need not itself be a rescaled version of the reference distribution.

The smooth-curve and two entropy reconstructions have 7,018 common valid-fit dates from the 8,825 premium dates. Requiring coverage of the fixed probability cutoffs leaves 6,966 dates, including 6,351 expansion dates and 615 recession dates. There are 394 usable adjacent-month comparisons from 438 potential comparisons and 129,733 eligible daily pairs from 176,432 potential pairs.

For each daily pair, label the date with the larger at-the-money premium h and the other date l . With $r = a_l/a_h$, common scale implies $G_h(k) = G_l(1 + r(k - 1))$. Daily pair comparisons are combined using product carry weights $D_t D_b$, normalized within each adjacent-month comparison. Summary statistics weight the monthly comparisons equally. Methods and cutoffs use the same eligible pairs. Calendar gaps are not bridged, and overlapping pairs are not independent observations.

Table A.21: Recovered-probability discrepancies from common scale

Method	K/F	RMS observed	RMS predicted	RMS discrepancy
Smooth curve	0.975	2.738	2.252	1.081
	0.950	2.802	2.310	0.775
	1.000	1.506	0.000	1.506
Entropy: lognormal	0.975	2.755	2.254	1.012
	0.950	3.158	2.415	0.963
	1.000	1.380	0.000	1.380
Entropy: uniform	0.975	2.663	2.242	0.908
	0.950	2.669	2.350	0.612
	1.000	1.409	0.000	1.409

Notes: Entries are percentage points. RMS denotes the root mean square across 394 adjacent-month comparisons. The first two columns of statistics refer to observed and scale-predicted probability changes; the final column refers to their difference. These are descriptive recovered-distribution comparisons, not quotation-based exclusions or sampling confidence intervals.

The separate quotation-bound calculation retains 6,999 usable dates in its expanded sample, adding 33 dates and one adjacent-month comparison to the main sample. All 2,367 monthly objectives in that calculation are numerically resolved. No signed-discrepancy outer interval excludes zero at $k = 0.95$, 0.975 , or 1 , in either sample. On the matched sample, median interval widths are 28.18, 32.57, and 35.13 percentage points, respectively. These intervals are too wide to establish a departure using the averaged discrepancy alone. Separate extrema may be attained by different distributions, so inclusion of zero does not establish that one distribution satisfies all restrictions jointly.

Fixed anchors and maturity aggregation. The selected put quotations include the 21 nearest at-the-money strikes and strict brackets around $K/F = 0.975$ and 0.95 , within the maintained strike window. Both sides of the forward enter the joint calculation.

At each component maturity, the premium anchor is fixed at the value obtained by strike interpolation of bid–ask midpoints. The original forward and carrying-cost inputs are also fixed. The component anchors are not optimized within the quotation bands. For the original maturity-interpolation weight w and carrying factors D_s, D_l , the daily mixture weights are

$$\alpha_s = \frac{(1 - w)D_s}{(1 - w)D_s + wD_l} \quad \alpha_l = \frac{wD_l}{(1 - w)D_s + wD_l}$$

The daily adjusted premium is $a_t = \alpha_s a_t^s + \alpha_l a_t^l$. Every component law retains unit mass, unit forward mean, its original premium anchor, and its selected quotation bands. Common scale applies to the daily mixture; it does not require identical laws at the two component maturities.

All exclusion statements condition on these fixed anchors, forwards, carrying costs, and quotation restrictions. The calculations do not assess sensitivity to allowing the anchors themselves to vary within their quotation bands.

Joint full-distribution scale calculation. Under common scale, $Z_t = (X_t - 1)/a_t$ has a common distribution across dates. Writing $C(x) = \mathbb{E}^\mathbb{Q}[(x - Z)^+]$ gives

$$\frac{\mathcal{P}_t(k)}{a_t} = C\left(\frac{k-1}{a_t}\right) \quad C(0) = 1$$

For a continuous distribution, $C'(0) = q$. More generally, the strict shortfall probability is the left derivative $C'_-(0)$; the right derivative also includes any atom at the forward. The common function is chosen once for the comparison window, rather than separately for each date.

For implementation, write $A = \max_t a_t$ within a window and transform returns to $R = 1 + (A/a_t)(X_t - 1)$. Quotation strikes transform by the same affine map. The linear program represents each component law using non-negative masses at the transformed payoff knots and an upper tail described by its mass and first moment. The relevant put payoffs are affine between knots, so this representation preserves their expectations. Equality of the weighted component masses and tail quantities across dates imposes the common daily law.

For clarity, let h index the positive-weight component maturities at date t , with weights α_{th} and anchors a_{th} . All quotations below are undiscounted put prices divided by the component forward. Set $\gamma_t = A/a_t$ and $\kappa_{thi} = 1 + \gamma_t(k_{thi} - 1)$ for quoted normalized strike k_{thi} . The common knot set contains zero, one, and every positive κ_{thi} ; write its largest knot as $\bar{r}$. For component (t, h) , let π_{thj} be mass at knot r_j , v_{th} upper-tail mass, and μ_{th} its first moment. The component moment restrictions are

$$\begin{aligned} \sum_j \pi_{thj} + v_{th} &= 1, & \sum_j r_j \pi_{thj} + \mu_{th} &= 1, \\ \sum_j (1 - r_j)^+ \pi_{thj} &= \gamma_t a_{th}, & \mu_{th} &\geq \bar{r} v_{th}. \end{aligned} \tag{A.28}$$

All masses and moments are nonnegative. One common daily law is imposed by requiring $\sum_h \alpha_{th} \pi_{thj} = v_j$, $\sum_h \alpha_{th} v_{th} = v_*$, and $\sum_h \alpha_{th} \mu_{th} = \mu_*$ at every date, with v_j, v_*, μ_* independent of t . The fitted original-unit quotation is $\hat{p}_{thi} = \gamma_t^{-1} \sum_j (\kappa_{thi} - r_j)^+ \pi_{thj}$. The objective is to minimize $\delta \geq 0$ subject to these restrictions and

$$p_{thi}^{\text{bid}} - \delta a_t \leq \hat{p}_{thi} \leq p_{thi}^{\text{ask}} + \delta a_t. \tag{A.29}$$

Thus δ is measured in fractions of the daily adjusted premium. The upper tail lies beyond every transformed quotation strike, so it has no put payoff at the retained strikes; its mass and moment preserve the remaining moment restrictions without a finite upper-price cap. For numerical implementation, one component at each date is eliminated using the mixture equalities and its nonnegativity is retained explicitly.

Allowing zero terminal prices and limiting tail configurations gives an outer closure of the class with strictly positive terminal equity values. Excluding this larger class excludes the maintained strict model. A fit in the outer class need not establish an attainable law with strictly positive terminal equity values.

When the original bands cannot be matched, the calculation permits a common allowance δa_t on each side of every selected band at date t , while keeping the premium anchors exact. A corrected dual lower bound above

10^{-8} certifies exclusion. The numerical primal solution supplies a fit and residual diagnostics; it is not a mathematically repaired upper certificate. Every full-scale window in the reported samples is numerically resolved. All exclusions in this run arise from the quotation restrictions rather than a standalone premium-anchor or support contradiction.

The magnitude of the pricing departures. Across the 405 main-sample monthly windows, the median corrected dual lower bound on the extra quotation allowance is 1.238 percent of the daily adjusted at-the-money premium. Among the 352 excluded months, the median is 1.393 percent. For adjacent-month windows, the corresponding all-window median is 2.126 percent.

To provide a spread benchmark, I interpolate each component maturity's normalized quoted half-spread to the forward using the same strike bracket as the midpoint anchor. Component half-spreads are expressed in the same units as their adjusted premiums and combined using the original carry-adjusted maturity weights. The median daily interpolated half-spread is 2.741 percent of the premium. The median of the within-month median half-spreads is 2.801 percent across all months and 2.717 percent among excluded months. Within the excluded months, the median ratio of the certified extra allowance to that month's median interpolated half-spread is 0.560.

The benchmark is an interpolated half-spread at the forward, not a directly quoted contract spread on every date or the spread of each binding wing quotation. The extra allowance is additional to the original bid-ask bands already admitted by the calculation. Its being smaller than a typical at-the-money half-spread therefore does not make the original restrictions compatible, or establish that every wing discrepancy is economically small. A certified lower bound is a minimum required allowance, not an allowance shown sufficient to restore compatibility. The reported bounds therefore do not supply an upper bound on the economic size of the discrepancy.

Conditional downside shape. The constant- q and varying- q models in (14) leave the upside law unrestricted across dates. Their comparison isolates the restriction that the broad shortfall probability remains constant. The full-scale model additionally restricts the upside shape.

Under the conditional restriction, $(1 - X_t) \mid X_t < 1 = m_t U$, where $\mathbb{E}[U] = 1$ and the distribution of U is common across dates. Since $m_t = a_t / q_t$, for $k \leq 1$,

$$\mathcal{P}_t(k) = q_t \mathbb{E}[(m_t U - (1 - k))^+] = a_t H\left(\frac{(1 - k)q_t}{a_t}\right)$$

Here $H(z) = \mathbb{E}[(U - z)^+]$, so $H(0) = 1$. The constant- q model chooses one unknown probability for the window; the varying- q model permits a separate probability at each date.

The first computational stage projects the unchanged component restrictions into marginal bounds on put prices at all selected downside strikes and at 0.95, 0.975, and 0.99, together with bounds on shortfall probability. The component grids are combined within each date, and the original carry-adjusted maturity weights give daily intervals. These separate marginal intervals enlarge the feasible set.

An inverse conditional put-curve formulation imposes necessary common-shape restrictions while solving for the unknown probabilities. A positive corrected certificate in this outer problem excludes every admissible probability path. Its objective is a numerical feasibility diagnostic, not an economically measured price deviation.

More explicitly, let $[L_t(k), U_t(k)]$ and $[q_t, \bar{q}_t]$ be the daily marginal projections. For $0 < k < 1$, intersect $[L_t(k)/a_t, U_t(k)/a_t]$ with $[0, k]$, using $0 \leq \mathcal{P}_t(k) \leq ka_t$, and write its endpoints as $\rho_t^-(k)$ and $\rho_t^+(k)$. Empty intersections are retained as infeasible constraints. Let $J = H^{-1}$, with $J(0)$ interpreted as the upper support endpoint. Its grid contains zero,

one, and all distinct price-interval endpoints. Necessary restrictions are

$$\begin{aligned}
 J(1) &= 0, \quad J \text{ is convex}, \quad \frac{J(\rho_{j+1}) - J(\rho_j)}{\rho_{j+1} - \rho_j} \leq -1, \\
 \max\{a_t, \underline{q}_t\} &\leq q_t \leq \min\{1, \bar{q}_t\}, \quad J(0) \leq \frac{q_t}{a_t}, \\
 J(\rho_t^+(k)) &\leq \frac{(1-k)q_t}{a_t}, \\
 \frac{(1-k)q_t}{a_t} &\leq J(\rho_t^-(k)) \quad \text{when } \rho_t^-(k) > 0.
 \end{aligned} \tag{A.30}$$

The last inequality is omitted at a zero lower price bound because $H \geq 0$ already supplies that bound. The slope restriction follows from $-1 \leq H' \leq 0$ on its decreasing part; convexity is preserved by inversion. Common conditional loss support and nonnegative terminal equity imply $J(0) \leq q_t/a_t$ at every date. These constraints are linear in the grid values of J and the unknown probabilities. The constant- q calculation uses one probability variable for the window, while the varying- q calculation uses one per date.

The implementation represents J in units of $1/\max_t a_t$, so its grid variables and probability variables lie in $[0, 1]$. Convexity is imposed without dividing by nearly coincident endpoint spacings; distinct floating-point endpoints are never merged. Each nonconstant constraint row is divided by the largest absolute coefficient or right-hand side; a constant-only row uses a divisor of at least one. A common nonnegative elastic slack, bounded by one, relaxes these normalized rows. Minimizing it gives an exclusion diagnostic for the marginal necessary system, rather than a price distance.

If the necessary problem is feasible, the candidate probabilities are passed to a second linear program restoring the complete original component quotation and moment restrictions. Only a numerically verified joint fit receives the outer-fit classification. A failed candidate remains unresolved because another probability vector may fit. The outer closure can include zero-loss endpoint mass, so candidate probabilities are not identified estimates of strict shortfall-event probabilities.

For a supplied probability path, this verification is a linear program. Set $m_t = a_t/q_t$ and $B = \min_t q_t/a_t$. Let v_j be a common unit-mass, unit-mean law on conditional-loss knots $u_j \in [0, B]$. For each component, its nonnegative labeled downside masses d_{thj} satisfy

$$\sum_h \alpha_{th} d_{thj} = q_t v_j, \quad m_t \sum_j u_j d_{thj} = a_{th}. \tag{A.31}$$

An otherwise unrestricted component upside law supplies the remaining mass and expected gain a_{th} . Its excess-return tail is represented by mass and first moment, with no finite upper cap. The knot grids retain every selected quotation payoff kink and the support endpoints. All original put bands, including strikes above the forward, are imposed jointly. The sparse stop-loss-curve implementation is equivalent to these mass restrictions; accepted solutions are converted back to masses and checked against the original component payoffs, moments, and mixture. A failed verification concerns only the supplied probability path and therefore cannot exclude the class with unknown q_t .

Numerical bounds and classification rules. The full-scale distance and inverse elastic programs have bounded variables. Write either in the form

$$\min_x c^T x \quad \text{subject to} \quad Bx \leq b, \quad Ex = d, \quad \ell \leq x \leq u. \tag{A.32}$$

For numerical inequality multipliers λ , replace positive entries by zero so that $\lambda \leq 0$, and let y denote the equality multipliers. Define $r = c - B^T \lambda - E^T y$. Weak duality gives the bounded-variable lower bound

$$L = b^T \lambda + d^T y + \sum_j \begin{cases} r_j \ell_j, & r_j \geq 0, \\ r_j u_j, & r_j < 0. \end{cases} \quad (\text{A.33})$$

The bound includes the adverse contribution of any residual reduced cost; it does not assume that numerical multipliers satisfy exact dual stationarity. In the inverse problem, the equality term is absent because $J(1) = 0$ is imposed as a variable bound. The implementation subtracts $10^{-10}(1 + S)$, where S is the sum of the absolute terms in L , rounds downward to the next representable floating-point value, and clips the lower bound at zero. Exclusion requires this corrected numerical lower bound to exceed 10^{-8} .

The solver's primal and dual feasibility tolerances are 10^{-9} . Accepted solutions have maximum inequality, equality, and variable-bound residual at most 5×10^{-8} in the implemented units. The full-scale calculation also checks the original component mass, mean, and anchor restrictions, and records remaining quotation slack. A joint conditional fit additionally requires maximum quotation slack at most 10^{-8} of the daily adjusted premium, with its mass, moment, and common-loss restrictions independently rechecked. The sparse interior-point verification uses an optimality tolerance of 10^{-10} . Nonfinite outputs, conflicting primal and dual diagnostics, failed solves, and time limits remain unresolved. A numerical witness is not an exact-arithmetic proof of attainability; exclusion certificates and fitted witnesses retain their distinct roles.

Nesting and propagation of existing fits. The reported classifications respect

$$\begin{aligned} \{\text{full common scale}\} &\subseteq \{\text{common conditional shape, constant } q\} \\ &\subseteq \{\text{common conditional shape, varying } q\}. \end{aligned}$$

This nesting also holds for the outer formulations used here.

To see the first inclusion, let a full-scale fit supply the common transformed law of $R = 1 + (A/a_t)(X_t - 1)$. Its premium satisfies $\mathbb{E}[(1 - R)^+] = A$. Let $q = \mathbb{Q}(R < 1)$ and, conditional on $R < 1$, define $U = q(1 - R)/A$. Then $\mathbb{E}[U] = 1$ and

$$1 - X_t = \frac{a_t}{A}(1 - R) = \frac{a_t}{q}U \quad \text{on } X_t < 1$$

The same conditional shape and the same q therefore apply at every date. The original component laws continue to satisfy their quotation, anchor, mass, and mean restrictions. Allowing the upside law to vary does not invalidate this fit. The second inclusion follows by allowing an already constant probability path in the varying- q class.

Verified fits from each smaller class are carried into the nested larger classes, retaining the accepted numerical residual standards. This changes no exclusion, quotation, anchor, or sample and does not turn numerical witnesses into exact-arithmetic certificates.

Coverage and compatibility across windows. Table A.22 reports the final window and sample comparisons, including inherited fits from nested classes. The expanded sample is selected using outer quotation feasibility and strike coverage. This selection is not itself a certificate of an attainable law with strictly positive terminal equity values.

Candidate failures and time limits are never counted as exclusions. The final main-sample classifications are 112 exclusions, 138 outer fits, and 155 unresolved months.

Table A.22: Coverage and joint compatibility across windows

Window/sample	Restriction	N	Excluded	Outer fit	Unresolved
<i>Main sample: 6,966 daily observations</i>					
Monthly	Full scale	405	352	53	0
	Conditional, constant q	405	345	55	5
	Conditional, varying q	405	112	138	155
Adjacent months	Full scale	394	389	5	0
	Conditional, constant q	394	388	5	1
	Conditional, varying q	394	174	6	214
<i>Main sample: at least ten dates per constituent month</i>					
Monthly	Full scale	356	319	37	0
	Conditional, constant q	356	312	39	5
	Conditional, varying q	356	90	116	150
Adjacent months	Full scale	348	343	5	0
	Conditional, constant q	348	342	5	1
	Conditional, varying q	348	133	5	210
<i>Expanded sample: 6,999 daily observations</i>					
Monthly	Full scale	406	354	52	0
	Conditional, constant q	406	347	54	5
	Conditional, varying q	406	116	133	157
Adjacent months	Full scale	395	390	5	0
	Conditional, constant q	395	389	5	1
	Conditional, varying q	395	179	6	210

Notes: Verified fits are inherited by nested larger model classes. No exclusion count changes through witness propagation. All calculations retain the original component quotations, fixed midpoint anchors, forwards, carrying costs, and maturity weights. An outer fit need not correspond to an attainable law with strictly positive terminal equity values. Unresolved outcomes remain in the denominators. Overlapping windows are separate compatibility exercises; the counts are not independent tests or sampling rejection frequencies.

What the nested comparisons establish. Of the 352 monthly full-scale exclusions, 345 remain excluded when the upside shape is unrestricted, two have a constant- q outer fit, and five remain unresolved. Thus most exclusions do not require imposing a common upside shape. This is not a downside-quotations-only test: the projections and joint verification retain all original component quotation and moment restrictions, including quotations above the forward.

Among the 345 constant- q exclusions, 80 have a verified joint outer fit when q_t varies, 112 remain excluded, and 153 remain unresolved. The probability paths in the 80 fits are witnesses of compatibility in the enlarged class, not estimates of the daily physical or strict priced probabilities.

The 112 exclusions establish that at least 27.7 percent of all 405 months require a departure from common conditional downside shape under the maintained measurement restrictions. For the outer class alone, assigning every unresolved month to incompatibility would raise the share to $(112 + 155)/405 = 65.9$ percent. This is not an upper bound for the class with strictly positive terminal equity values, because outer fits need not be attainable there. Neither endpoint is a sampling confidence bound.

Direction of departures and unresolved windows. The saved outputs contain corrected scalar dual bounds and primal quotation fits, but not the full vectors of dual multipliers. Quotations attaining the maximum allowance in a saved minimax fit need not form a unique conflicting set, and an active quotation can have zero dual weight. They therefore do not identify a unique directional tail response.

Across the 352 excluded main-sample months, the saved fits contain binding downside quotations both above asks and below bids. They also contain binding quotations on dates both above and below the within-window

median premium. These diagnostics do not support a uniform direction of departure in the left tail.

The unresolved months are not confined to the earliest or sparsest observations. Their median coverage is 19 dates and 829 selected component quotations, compared with 20 dates and 878 quotations in the fitted months. Of the 155 unresolved months, 150 contain at least ten retained dates. The years 2000–2004 account for 45 unresolved months out of 60 covered months. These classifications describe limits of the numerical feasibility search, not low statistical test power.